University of Amsterdam

MSc Stochastics and Financial Mathematics

Master Thesis

Optimization with Quantum Annealing

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Examination date:
February, 2026

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Abstract

This thesis investigates the feasibility of implementing mean-expected shortfall (Mean-ES) portfolio optimization on quantum annealing hardware, presents the mathematical foundations of quantum mechanics within the Dirac-von Neumann formalism, and exhibits the theory of quantum annealing with the adiabatic theorem and the construction of problem Hamiltonians from Quadratic Unconstrained Binary Optimization (QUBO) problems.

While mean–variance optimization fits naturally on quantum annealers, expected shortfall (ES) presents challenges. ES admits a piecewise linear reformulation compatible with QUBO through binary encoding. Motivated by the Basel III regulatory shift from Value-at-Risk to ES, the thesis translates the Mean-ES optimization into QUBO formulation. Contributions include: (1) derivation of portfolio-independent bounds for auxiliary variables using Cantelli's inequality and the Rayleigh quotient; (2) application of two approaches for handling inequality constraints - slack variables for exact satisfaction and unbalanced penalization for reduced qubit usage; (3) partial demonstration of feasibility on D-Wave's quantum annealer with S&P 500 data, though validation remains for future work. To our knowledge, this represents the first attempt at translating a coherent, regulatory-compliant risk measure to quantum annealing hardware directly via the QUBO formulation.

Keywords: Quantum annealing, portfolio optimization, expected shortfall
MSC: 91G10, 81P68

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Contents

Notation **5**

Introduction **7**

1 Introduction to quantum mechanics **11**
1.1 Wave function . . . 11
1.2 Dirac-Von Neumann Formalism . . . 13
1.2.1 Hilbert spaces . . . 13
1.2.2 Orthonormal bases . . . 15
1.2.3 Riesz representation theorem . . . 16
1.2.4 Operators on Hilbert spaces . . . 17
1.2.5 Dirac bra-ket notation . . . 18
1.2.6 Spectral decomposition . . . 19
1.2.7 Quantum states . . . 22
1.2.8 Composite systems . . . 26
1.2.9 Measurement . . . 26
1.2.10 The Schrödinger equation . . . 27

2 Quantum annealing **28**
2.1 The quantum adiabatic theorem . . . 28
2.1.1 Time-dependent Hamiltonians and instantaneous eigenstates . . . 28
2.1.2 Introduction to adiabaticity . . . 29
2.1.3 Differentiation of operators on finite dimensional Hilbert spaces . . . 31
2.1.4 Hamiltonian interpolation . . . 32
2.1.5 Adiabatic theorem . . . 32
2.2 Optimization with quantum annealing . . . 33
2.2.1 Quantum annealer . . . 34
2.2.2 Quadratic unconstrained binary optimization (QUBO) . . . 34
2.2.3 Initial Hamiltonian . . . 36
2.2.4 Construction of the problem Hamiltonian H_P . . . 37
2.2.5 Proof of Theorem 2.6 . . . 38
2.2.6 Quantum annealing procedure . . . 41

3 Portfolio optimization **43**
3.1 Static portfolio . . . 43
3.2 Risk measures . . . 44
3.3 Value at risk (VaR) . . . 45
3.4 Expected shortfall (ES) . . . 45

3.5 Coherent risk measures . . . 48
3.6 Mean-ES portfolio . . . 49
3.7 Budgetary constraint . . . 50

4 QUBO Mean-ES portfolio **51**
4.1 Binary encoding . . . 51
4.2 Binary encoding decision variables . . . 52
4.2.1 Binary portfolio . . . 52
4.2.2 Binary encoding τ . . . 53
4.2.3 Binary encoding y_t . . . 55
4.3 Unconstraining the Mean-ES portfolio . . . 55
4.3.1 Slack-variables approach . . . 56
4.3.2 Unbalanced-penalization approach . . . 57

5 Case study **59**
5.1 Dataset . . . 59
5.2 Implementation . . . 59
5.3 Results . . . 60

6 Discussion and future work **61**

Popular Summary **64**

Bibliography **66**

Notation

Symbol	Meaning / definition
General sets, scalars, vectors	
$\mathbb{N}, \mathbb{R}, \mathbb{C}$	Natural, real, and complex numbers.
i	Imaginary unit in $\mathbb{C}$, characterized by $\mathrm{i}^2 = -1$.
$\overline{z}$	Complex conjugate of $z \in \mathbb{C}$.
$\mathbb{1}_A$	Indicator of a set A: $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise.
$\mathscr{B}(\mathbb{R})$	Borel σ-algebra on $\mathbb{R}$
$x \cdot y$	Euclidean (ordinary) inner product on $\mathbb{R}^d$: $x \cdot y := \sum_{k=1}^{d} x_k y_k$.
$\lVert \cdot \rVert$	Norm induced by an inner product on a Hilbert space (see below) or, for operators, the operator norm (see below).
Hilbert spaces and operators	
$\mathcal{H}$	A (real or complex) Hilbert space.
$\langle \cdot \mid \cdot \rangle$	Inner product on $\mathcal{H}$ (Dirac bra–ket notation). The induced norm is $\lVert\psi\rVert_{\mathcal{H}} := \sqrt{\langle\psi\mid\psi\rangle}$.
$\lvert\psi\rangle,\ \langle\phi\rvert$	Ket (vector in $\mathcal{H}$) and bra (continuous linear functional on $\mathcal{H}$). The inner product is $\langle\phi\mid\psi\rangle$.
$A : D(A) \to \mathcal{H}$	(Possibly unbounded) linear operator with domain $D(A) \subseteq \mathcal{H}$.
A^*	Adjoint of A. A is *self-adjoint* if $A = A^*$ and $D(A) = D(A^*)$.
$\lVert A \rVert$	Operator norm for a bounded operator A: $\lVert A\rVert := \sup_{\psi\neq 0} \frac{\lVert A\psi\rVert}{\lVert\psi\rVert}$.
$\mathcal{B}(\mathcal{H})$	Bounded linear operators on $\mathcal{H}$.
$\mathrm{Tr}(\cdot)$	Trace of an operator.
I	Identity operator (or identity matrix, in finite dimension).
$\sigma(A)$	Spectrum of an operator A.
Spectral measures and measurement	
(S, Σ)	Measurable space.
$P : \Sigma \to \mathcal{B}(\mathcal{H})$	Projection-valued measure (PVM).
P_A	PVM associated with a self-adjoint operator A (spectral theorem).
μ_A	Probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ induced by observable A and state ρ via the Born–von Neumann formula $\mu_A(\Omega) = \mathrm{Tr}(P_A(\Omega)\rho)$.
Quantum-mechanical constants and dynamics	
h	Planck's constant.
$\hbar$	Reduced Planck constant: $\hbar := \frac{h}{2\pi}$.
$H(t)$	(Time-dependent) Hamiltonian operator governing dynamics of the system.
T_f	Total evolution time (final time) in annealing/adiabatic evolution.
$E_j(t),\ \lvert\varphi_j(t)\rangle$	Instantaneous eigenvalues and eigenstates of $H(t)$: $H(t)\lvert\varphi_j(t)\rangle = E_j(t)\lvert\varphi_j(t)\rangle$.
$\Delta(t)$	Instantaneous spectral gap (for nondegenerate ground state): $\Delta(t) := E_2(t) - E_1(t)$.
X, Y, Z	Pauli matrices on $\mathbb{C}^2$.

Continued on next page

Symbol	Meaning / definition
$\otimes$	Tensor product. For n qubits, $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$.
Z_i	Pauli-Z acting on qubit i: $Z_i := I^{\otimes(i-1)} \otimes Z \otimes I^{\otimes(n-i)}$.
Optimization and portfolio-risk notation	
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space.
$\mathbb{E}[\cdot]$	Expectation under $\mathbb{P}$.
$\mathrm{Var}(\cdot),\ \mathrm{Cov}(\cdot)$	Variance and covariance.
$\lambda_{\max}(\Sigma)$	Largest eigenvalue of a covariance matrix Σ.
$\bar{\pi} = (\pi_0, \pi),\ \bar{S} = (S_0, S)$	Asset price vectors at times $t = 0$ and $t = 1$, with risk-free asset indexed by 0.
$\bar{\xi} = (\xi_0, \xi)$	Portfolio holdings (risk-free and risky components).
r	Risk-free rate; $S_0 = \pi_0(1 + r)$ with the convention $\pi_0 = 1$.
$\bar{R} = (r, R)$	Return vector (risk-free and risky).
W_0, W_1	Portfolio value at times $t = 0$ and $t = 1$: $W_0 = \bar{\xi} \cdot \bar{\pi}$, $W_1 = \bar{\xi} \cdot \bar{S}$.
L	Portfolio loss: $L := W_0 - W_1 = -\bar{\xi} \cdot \bar{R}$.
F_L	Distribution function of L: $F_L(x) := \mathbb{P}(L \leqslant x)$.
$F_L^{\leftarrow}$	Generalized inverse: $F_L^{\leftarrow}(u) := \inf\{x \in \mathbb{R} : F_L(x) \geqslant u\}$.
$\mathrm{VaR}_\alpha(L)$	Value-at-Risk at level $\alpha \in (0, 1)$: $\mathrm{VaR}_\alpha(L) = F_L^{\leftarrow}(\alpha)$.
$\mathrm{ES}_\alpha(L)$	Expected shortfall at level $\alpha \in (0, 1)$: $\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 F_L^{\leftarrow}(u)\, \mathrm{d}u$.
$(x)^+$	Positive part: $(x)^+ := \max(0, x)$.
α	Confidence level in $(0, 1)$ for VaR_α and ES_α.
τ	Auxiliary (threshold) variable in the Rockafellar–Uryasev reformulation; at optimum, $\tau = \mathrm{VaR}_\alpha(L)$.
λ	Trade-off parameter weighting expected return against risk in mean–ES formulations.
y_t	Auxiliary variables used to represent positive parts/inequalities in sample-based ES formulations (e.g. $y_t \geqslant 0$ and $y_t \geqslant -\bar{\xi} \cdot \bar{r}_t - \tau$).
n	Number of binary variables, and number of (logical) qubits in QUBO formulation.
$x \in \{0, 1\}^n$	Binary decision vector in QUBO.
Q	Real $n \times n$ coefficient matrix in a QUBO model (often taken upper triangular in this thesis).
$f(x)$	QUBO objective $f(x) = x^\mathsf{T} Q x = \sum_i Q_{ii} x_i + \sum_{i<j} Q_{ij} x_i x_j$.
H_P	Problem Hamiltonian encoding an objective f via $H_P\lvert x\rangle = f(x)\lvert x\rangle$ on the standard basis.
B	Budget parameter (e.g. $\bar{\xi} \cdot \bar{\pi} = B$).
T	Sample size (number of return observations) used in empirical estimators.

Introduction

The mathematical description of physical reality underwent a profound transformation in the early twentieth century with the development of quantum mechanics. Schrödinger (1926), Dirac (1930), and Neumann (1932) formalized the quantum mechanical theory that replaced deterministic classical trajectories with probabilistic wave functions evolving in complex Hilbert spaces. The Schrödinger equation

$$\mathrm{i}\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi$$

governs the time evolution of quantum states, where $\hbar$ is the reduced Planck constant, Ψ is the wave function representing the quantum state, and $\hat{H}$ is the Hamiltonian operator encoding the system's total energy. Researchers developed this mathematical framework to explain atomic and subatomic phenomena, and they have recently applied it to optimization through the paradigm of quantum annealing.

The rigorous mathematical foundations of quantum mechanics rest upon the Dirac-von Neumann formalism, which associates quantum systems with complex Hilbert spaces and observables with self-adjoint operators. The spectral theorem guarantees that self-adjoint operators possess real spectra, ensuring that physical measurements yield real values. For finite-dimensional systems, the Dirac-von Neumann formalism reduces to linear algebra over complex vector spaces, with quantum states represented as unit vectors and observables as Hermitian matrices. The tensor product structure of composite quantum systems implies that an n-qubit system resides in the Hilbert space

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes n},$$

which has dimension 2^n. Vazirani (1998) and Jordan (2008) point this exponential growth of the state space with only n qubits as part of the explanation of the power of quantum computation.

The quantum adiabatic theorem provides the theoretical foundation for quantum annealing. Born and Fock (1928) rigorously formulated the quantum adiabatic theorem that establishes conditions under which a quantum system initialized in the ground state of a time-dependent Hamiltonian $H(t)$ remains close to the instantaneous ground state throughout its evolution. Specifically, if the Hamiltonian interpolates between an initial Hamiltonian H_I with an easily prepared ground state and a problem Hamiltonian H_P whose ground state encodes the solution to an optimization problem, then sufficiently slow evolution ensures the final measured state approximates the desired solution. The evolution time scales inversely with the power of the spectral gap

$$\Delta(t) := E_2(t) - E_1(t)$$

between the ground and first excited states, making the spectral gap analysis crucial for understanding algorithmic performance.

The practical implementation of quantum annealing requires translating optimization problems into a specific mathematical form. Quantum annealers, such as those developed by D-Wave Quantum Inc. (2025b), solve Quadratic unconstrained binary optimization (QUBO) problems,

$$\min_{x \in \{0,1\}^n} x^T Q x,$$

where the matrix $Q \in \mathbb{R}^{n \times n}$. The construction of the problem Hamiltonian H_P from a QUBO objective function proceeds through a mapping using the so-called Pauli-Z operators. This construction ensures that the eigenvalues of H_P coincide with the objective function values, and the ground state corresponds to the global minimum.

Markowitz (1952) introduced mean-variance optimization to balance risk and return in portfolio optimization. The quadratic nature of mean-variance optimization fits well on the emerging quantum annealing hardware. Several researchers have successfully implemented Markowitz portfolio selection on quantum annealers. Elsokkary et al. (2017) optimized a portfolio of 63 securities on D-Wave's 128-qubit processor. Cohen and Alexander (2020) combined classical and quantum solvers for portfolio construction from 3171 stocks. Venturelli and Kondratyev (2019) adopted reverse annealing to optimize risk-adjusted returns using the Sharpe ratio. TNO researchers Aguilera et al. (2024) and Phillipson and Bhatia (2020) incorporated multi-objective optimization with environmental constraints, concentration risk measures, and emission targets aligned with the Paris Agreement. This thesis explores the critical question of whether risk measures beyond variance fit quantum annealing. The challenge is non-trivial because quantum annealers require problem reformulation as the QUBO form. This constraint immediately eliminates many sophisticated risk measures from consideration.

An extensive survey of risk measures reveals significant limitations for quantum implementation. Path-dependent measures like maximum drawdown require temporal information incompatible with the static QUBO formulation. Higher-order polynomial risk measures such as skewness and kurtosis exceed the quadratic structure. Expected Shortfall (ES), also known as Conditional Value at Risk (CVaR), emerges as an exception because Rockafellar and Uryasev (2000) proved the crucial reformulation

$$\mathrm{ES}_\alpha(L) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{1-\alpha} \mathbb{E}\left[(L-\tau)^+\right] \right\}.$$

This representation is linear in the auxiliary variable τ and involves only the piecewise linear function $(L-\tau)^+ = \max(0, L-\tau)$. This reformulation approximately fits the QUBO model through binary encoding and constraint handling.

The selection of ES for quantum implementation is coinciding with current regulatory developments. The Basel Committee on Banking Supervision mandated a transition from VaR to ES as the primary risk measure for internal models under Basel III (Basel 2025) following the 2007-2008 financial crisis. ES captures the severity of losses in the tail of the loss distribution, while VaR gives little information about how large losses are in the tail. Capturing the losses is the tail is important to understand the losses

in the most severe cases. The Basel committee specified a precise calculation of the parameters. In particular, the internal model evaluates ES at 97.5% confidence level over 10-day horizon using 12 months of historical data (Basel 2025). ES satisfies all coherence axioms (Artzner et al. 1999), unlike VaR, which suffers from non-subadditivity and can penalize diversification.

The central mathematical challenge lies in reformulating the Mean-ES portfolio problem as QUBO while maintaining computational tractability on current quantum hardware. This requires three critical transformations. First, continuous portfolio weights $\xi \in [a, b]$ are binary encoded via

$$\xi^* = a + \frac{b-a}{2^K - 1} \sum_{k=0}^{K-1} 2^k b_k$$

with binary variables $b_k \in \{0, 1\}$, and approximation error $|\xi - \xi^*| \leqslant \frac{b-a}{2(2^K-1)}$. Second, inequality constraints require either slack variables or penalty functions with trade-offs between qubit usage and solution quality. Third, efficient encoding requires establishing portfolio-independent bounds for all auxiliary variables.

A key theoretical contribution is proving that the optimal $\tau = \mathrm{VaR}_\alpha(L)$ satisfies

$$\tau \in \left[-rB + \sum_{i=1}^{d} \mathbf{1}_{\{E[R_i] > r\pi_i\}} (r\pi_i - E[R_i]) - \sqrt{\frac{1-\alpha}{\alpha}} \sqrt{\lambda_{\max}(\Sigma)(B^2 + d)},\ B \right].$$

This bound (See Proposition 4.2), derived using Cantelli's inequality and the Rayleigh quotient, enables binary encoding with quantifiable approximation guarantees, which is a critical requirement for practical quantum implementation.

The thesis makes three principal contributions. First, we develop a mathematical framework for translating Mean-ES portfolio optimization into QUBO formulation, including the derivation of portfolio-independent bounds for the auxiliary variable τ using Cantelli's inequality and the Rayleigh quotient. Second, we present two approaches for handling the inequality constraints arising from the ES reformulation: a slack-variable method that maintains exact constraint satisfaction at the cost of additional qubits, and an unbalanced-penalization approach following Montañez-Barrera et al. (2024) that reduces qubit requirements. Third, we provide a partial demonstration of practical feasibility by implementing the QUBO model on D-Wave's quantum annealer using S&P 500 data under Basel III parameters, though full validation and parameter calibration remain for future work.

To our knowledge, this work represents the first attempt at directly translating a coherent, regulatory-compliant risk measure to quantum annealing hardware. While the current implementation requires further refinement, the framework developed here may serve as a starting point for exploring quantum-enhanced portfolio optimization beyond variance-based models.

The remainder of this thesis is organized as follows. Chapter 1 establishes the quantum mechanical foundations necessary for understanding quantum annealing, beginning with the wave function formalism and proceeding through the Dirac–von Neumann axioms of

Hilbert spaces, self-adjoint operators, the spectral theorem, quantum states, composite systems, and measurement. The chapter provides the mathematical prerequisites for formalizing quantum evolution via the Schrödinger equation. Chapter 2 presents the quantum annealing algorithm and the adiabatic theorem, including the construction of problem Hamiltonians from QUBO formulations and the quantum annealing procedure. Chapter 3 develops the portfolio optimization framework focusing on ES, its characterization as a coherent risk measure, and the mean-ES portfolio optimization problem. Chapter 4 contains the main technical contribution: the QUBO reformulation of Mean-ES portfolios, including binary encoding schemes and constraint handling approaches. Chapter 5 validates the approach in this thesis through a partial implementation on D-Wave hardware using S&P 500 data.

1 Introduction to quantum mechanics

This chapter provides the necessary basics of quantum mechanics for the mathematical formalization of quantum annealing. Section 1.1 recalls the formalization of quantum mechanics via wave functions and the time-dependent Schrödinger equation; Sections 1.2.1–1.2.6 introduce more rigorous Dirac–von Neumann formalism (Hilbert spaces, observables as self-adjoint operators, spectral theorem) for quantum mechanics. Sections 1.2.7–1.2.9 formalize states and measurement; Section 1.2.10 closes with the time evolution of quantum states.

1.1 Wave function

Quantum mechanics provides us with the probability of finding a particle in a specific location. We can calculate this probability using the particle's wave function $\Psi(\mathbf{r}, t)$, which we obtain by solving the Schrödinger equation.

The time-dependent Schrödinger equation (Schrödinger 1926) is

$$\mathrm{i}\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi, \tag{1.1}$$

with the wave function

$$\Psi : \mathbb{R}^d \times \mathbb{R}_+ \longrightarrow \mathbb{C},$$

where $d \in \mathbb{N}$, i is the imaginary unit, $\hbar = \frac{h}{2\pi}$ is the Planck's constant h divided by 2π, and $\hat{H}$ is the Hamiltonian operator. Given the particle's initial wave function $\Psi(\mathbf{r}, 0)$, the Schrödinger equation determines the wave function $\Psi(\mathbf{r}, t)$ for all future times.

For $d = 3$, the wave function $\Psi(\mathbf{r}, t)$ is complex-valued and represents the state of a particle at a point $\mathbf{r} = (x, y, z) \in \mathbb{R}^3$ and time t. Born's statistical interpretation (Born 1926) says that the probability of finding the particle in volume A at time t is

$$\int_A |\Psi(\mathbf{r}, t)|^2 \,\mathrm{d}^3\mathbf{r}.$$

The wave function Ψ is normalized,

$$\int_{\mathbb{R}^3} |\Psi(\mathbf{r}, t)|^2 \,\mathrm{d}^3\mathbf{r} = 1,$$

to describe a possible physical state. Figure 1.1 shows an example of a wave function of one particle in a one-dimensional space.

Quantum mechanics cannot predict the exact position of a particle with certainty. While we can measure the particle's position, we cannot predict with certainty the

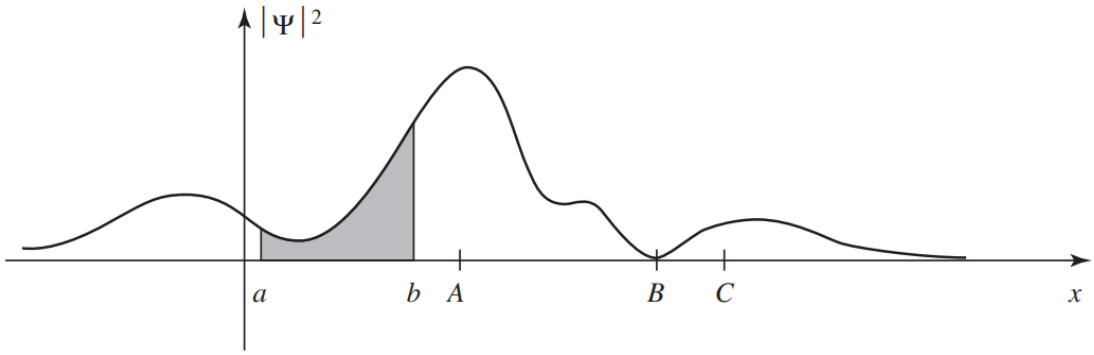

Figure 1.1: The plot (Griffiths and Schroeter 2018) shows a typical wave function. The shaded area represents the probability of finding the particle between a and b in a one-dimensional space. The particle is relatively likely to be found near point A, and unlikely to be found near point B.

outcome of the measurement. The so-called Copenhagen interpretation (Mermin 1985) is that the measurement compels the particle to assume a definite position.

Suppose the particle assumes position C in Figure 1.1 upon measurement. We say that the wave function collapses (Griffiths and Schroeter 2018) upon measurement to a spike at the point C. The wave function soon spreads out again according to the Schrödinger equation.

The Hamiltonian operator $\hat{H}$ represents the system's total kinetic and potential energy. An operator is a mathematical entity that acts on a quantum state to produce another quantum state. The Hamiltonian operator $\hat{H}$ governs the evolution of the quantum state by acting on the wavefunction $\Psi(\mathbf{r}, t)$ via the Schrödinger equation (1.1) and producing the next state.

The expression of the Hamiltonian operator depends on the specificities of the system under consideration. For example, one particle in a three-dimensional space (Rashid 2020, Equation (16)) has the Hamiltonian operator

$$\begin{aligned}\hat{H} &= \hat{K} + \hat{V} \\ &= -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t)\end{aligned}$$

where m is the mass of the particle, $\hat{K}$ is the kinetic energy operator, $\hat{V} = V(\mathbf{r}, t)$ is the potential energy operator, and $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplacian. The particle is moving in an external potential $V(\mathbf{r}, t)$. Notice that the potential operator $V(\mathbf{r}, t)$ simply multiplies the wavefunction $\Psi(\mathbf{r}, t)$.

We may consider larger and more complicated systems, such as an atom surrounded by Z electrons. The wave function $\Psi(\mathbf{r}_1, ..., \mathbf{r}_Z, t)$ describes the state of the system. The state of the system is a function of time t, the coordinate $\mathbf{r}_1$ of the first electron, the coordinate $\mathbf{r}_2$ of the second electron, etc. The time evolution of the system follows the Schrödinger equation.

1.2 Dirac-Von Neumann Formalism

Von Neumann (1932) together with Dirac (1930) stated the first complete mathematical formulation of quantum mechanics. Their axiomatic mathematical formalism describes quantum mechanics rigorously using Hilbert spaces and operators on these spaces.

1.2.1 Hilbert spaces

Let $\mathbb{K}$ denote the real field $\mathbb{R}$ or complex field $\mathbb{C}$.

Definition 1.1. A Hilbert space $\mathcal{H}$ is a vector space over the field $\mathbb{K}$ with an inner product $\langle\cdot|\cdot\rangle : \mathcal{H} \times \mathcal{H} \longrightarrow \mathbb{K}$ such that relative to the metric $d(x, y) := ||x - y||_{\mathcal{H}}$ induced by the norm $|| \cdot ||_{\mathcal{H}} := \sqrt{\langle\cdot|\cdot\rangle}$, $\mathcal{H}$ is a complete metric space.

Every Cauchy sequence of vectors in the Hilbert space $\mathcal{H}$ converges to a vector in the Hilbert space $\mathcal{H}$ by definition.

Mathematicians denote the inner product on a Hilbert space $\mathcal{H}$ by $\langle\cdot,\cdot\rangle$ or $\langle\cdot,\cdot\rangle_{\mathcal{H}}$. Physicists use the Dirac bra-ket notation $\langle\cdot|\cdot\rangle$ or $\langle\cdot|\cdot\rangle_{\mathcal{H}}$. These two notations relate as

$$\langle x, y\rangle = \langle y|x\rangle \quad \text{for all } x, y \in \mathcal{H}.$$

The first axiom in the Dirac-Von Neumann formalism associates a quantum system with a Hilbert space. The physics of the system determines the dimensionality of the Hilbert space.

Axiom 1 (Postulate I)**.** *The state of an isolated (pure) physical system is represented, at a fixed time t, by a unit vector $|\psi(t)\rangle$ belonging to a complex Hilbert space $\mathcal{H}$ called the state space. If unit vectors $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$ in the complex Hilbert space $\mathcal{H}$ satisfy $|\psi_1(t)\rangle = e^{\mathrm{i}\theta}|\psi_2(t)\rangle$ for a phase $\theta \in \mathbb{R}$, then the vectors $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$ represent the same quantum state.*

Some formulations additionally state that the Hilbert space $\mathcal{H}$ is topologically separable and complex.

Postulate I states that a quantum state is properly represented by an equivalence class $[|\psi(t)\rangle]$ of unit vectors in a complex Hilbert space $\mathcal{H}$. A unit vector $|\psi_1(t)\rangle \sim |\psi_2(t)\rangle$ if and only if $|\psi_1(t)\rangle = e^{\mathrm{i}\theta}|\psi_2(t)\rangle$ for a phase $\theta \in \mathbb{R}$. The phase factor $e^{\mathrm{i}\theta}$ is a unit complex number, and the unit vectors $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$ differing only by a phase factor $e^{\mathrm{i}\theta}$ represent the same quantum state.

The definition of a quantum system seems to be somewhat elusive. For example, Peres (2002, Section 2-1.) "defines" the quantum system as whatever admits a closed dynamical description within quantum theory, and argues that one must be content with such vagueness. Examples of quantum systems are hydrogen atom, hydrogen atom with a photon, or the sole magnetic momentum the silver atom.

Notice that Postulate I uses the Dirac bra-ket notation for the vector $|\psi(t)\rangle \in \mathcal{H}$. This thesis uses both classical and bra-ket notation for vectors, which may be confusing. The reason is to familiarize oneself with the widely-spread and efficient bra-ket notation, but

to avoid awkward formulations of bra-ket notation. For example, physicists simplify the bra-ket inner product notation as

$$\langle |\phi\rangle || \psi\rangle \rangle = \langle \phi | \psi \rangle.$$

Subsequent sections further explain and formalize the Dirac bra-ket notation.

The following two examples are important Hilbert spaces in quantum mechanics and quantum computing.

Example 1 ($\mathbb{C}^d$)**.** *The space $\mathbb{C}^d$, for $d \in \mathbb{N}$, is finite-dimensional Hilbert space with respect to the inner product*

$$\langle \phi | \psi \rangle = \sum_{i=1}^{d} \overline{\phi}_i \, \psi_i.$$

In particular, a qubit is a quantum system associated with the Hilbert space $\mathcal{H} = \mathbb{C}^2$. The quantum system qudit is associated with a Hilbert space $\mathcal{H} = \mathbb{C}^d$, where $d \in \mathbb{N}$.

Definition 1.2 ($\mathcal{L}^2$-space)**.** Let (S, Σ, μ) be a measure space and let $\mathbb{K}$ denote the real or complex field. Define the function space

$$\mathcal{L}^2(S, \Sigma, \mu) \coloneqq \left\{ f : S \to \mathbb{K} \;\; \Sigma\text{-measurable} : \; \int_S |f|^2 \, \mathrm{d}\mu < \infty \right\}.$$

The function space $\mathcal{L}^2(S, \Sigma, \mu)$ with respect to the map

$$\langle f | g \rangle \coloneqq \int_S \overline{f(s)} \, g(s) \, \mathrm{d}\mu(s) \tag{1.2}$$

is not always a Hilbert space because the map is not necessarily an inner product. Any inner product $\langle \cdot | \cdot \rangle$ must be positive-definite by definition. An inner product is positive-definite if

$$\langle f | f \rangle > 0$$

for every nonzero function f.

Consider the measure space $(S, 2^S, \mu)$ with $S = \{a, b\}$. Set $\mu(\{a\}) = 0$ and $\mu(\{b\}) = 1$. Let $f \coloneqq \mathbb{1}_{\{a\}}$. The map (1.2) is not positive-definite because nonzero $f \in \mathcal{L}^2(S, \Sigma, \mu)$ gives

$$\begin{aligned} \langle f | f \rangle &= \int_S \overline{f(s)} \, f(s) \, \mathrm{d}\mu(s) \\ &= \int_{\{a\}} \mathbb{1}_{\{a\}} \mathrm{d}\mu(s) \\ &= \mu(\{a\}) = 0. \end{aligned}$$

This example shows that not every $\mathcal{L}^2(S, \Sigma, \mu)$ space with the inner product (1.2) is a Hilbert space. The non-Hilbert spaces $\mathcal{L}^2(S, \Sigma, \mu)$ do not associate with quantum systems under Postulate I (Axiom 1), but are crucial for defining Lebesgue spaces $L^2(S, \Sigma, \mu)$. Lebesgue spaces $L^2(S, \Sigma, \mu)$ are Hilbert spaces and are associated with quantum systems under Postulate I.

Definition 1.3 (The L^2-space)**.** Let (S, Σ, μ) be a measure space and let $\mathbb{K}$ denote the real or complex field. Let

$$\mathcal{L}^2(S, \Sigma, \mu) \coloneqq \left\{ f : S \to \mathbb{K} \;\; \Sigma\text{-measurable} : \int_S |f|^2 \, \mathrm{d}\mu < \infty \right\}.$$

Define an equivalence relation on $\mathcal{L}^2(S, \Sigma, \mu)$ by

$$f \sim g \quad \Longleftrightarrow \quad f = g \;\; \mu\text{-a.e.}$$

The $L^2(S, \Sigma, \mu)$ space is the quotient vector space

$$L^2(S, \Sigma, \mu) \coloneqq \mathcal{L}^2(S, \Sigma, \mu)/\sim,$$

whose elements are equivalence classes $[f]$ of functions. On $L^2(S, \Sigma, \mu)$ define map

$$\langle [f] | [g] \rangle \coloneqq \int_S \overline{f} \, g \, \mathrm{d}\mu.$$

The map is well-defined (independent of representatives), $\langle \cdot | \cdot \rangle$ is an inner product, and $(L^2(S, \Sigma, \mu), \langle \cdot | \cdot \rangle)$ is a Hilbert space. The $L^2(S, \Sigma, \mu)$ is complete by the Riesz–Fischer theorem (Fischer 1907).

Some wave functions live in the L^2 spaces.

Example 2. *The wavefunctions describing the quantum state of an electron in the hydrogen atom live in the $L^2(\mathbb{R}^3)$ space (Fewster 1993, Chapter 2).*

1.2.2 Orthonormal bases

Orthonormal bases are fundamental for studying quantum mechanical systems. Every Hilbert space has an orthonormal basis (Tian and Jorgensen 2010, Theorem 1.9).

Definition 1.4. Let I be an index set. A set $\{e_i\}_{i \in I}$ of elements of a Hilbert space $\mathcal{H}$ is called orthogonal if

$$\langle e_i | e_j \rangle = 0 \quad \text{for all } i \neq j.$$

If additionally $\|e_i\| = 1$ for all $i \in I$, the set $\{e_i\}_{i \in I}$ is called orthonormal. An orthonormal set $\{e_i\}_{i \in I}$ is called an orthonormal basis for $\mathcal{H}$ if the space of finite linear combinations of the e_j's is dense in $\mathcal{H}$.

An infinite-dimensional Hilbert space has a countable orthonormal basis if the Hilbert space is separable.

Definition 1.5. A normed space $(N, \|\cdot\|)$ is called separable if there exists a countable dense subset of N.

Theorem 1.1. *Every Hilbert space $\mathcal{H}$ has an orthonormal basis. If $\mathcal{H}$ is infinite-dimensional and separable, then there exists an orthonormal basis $(e_n)_{n \in \mathbb{N}}$ of $\mathcal{H}$.*

Proof. By Sunder (2014, Proposition 1.4.8). □

The orthonormal basis allows the (pure) quantum states in the Hilbert space to be expressed in terms of the basis elements.

Theorem 1.2. *Let I be an index set. Suppose $\{e_j\}_{j\in I}$ is an orthonormal basis for a Hilbert space $\mathcal{H}$. Every $\psi \in \mathcal{H}$ can be expressed uniquely as a convergent sum*

$$\psi = \sum_{j\in I} a_j e_j,$$

where the coefficients

$$a_j = \langle \psi | e_j \rangle.$$

Proof. By Conway (1990, Theorem 4.13(d), Lemma 4.12). □

1.2.3 Riesz representation theorem

Definition 1.6. Let $\mathbb{K}$ be the real or complex field. The topological dual space of a Hilbert space $\mathcal{H}$ is defined as

$$\mathcal{H}' := \{\varphi : \mathcal{H} \to \mathbb{K} : \varphi \text{ linear and continuous}\}.$$

The space $\mathcal{H}'$ is called the topological dual space of $\mathcal{H}$ because its elements are continuous. The so-called algebraic dual space consists of all linear functionals.

Definition 1.7. Let $\mathcal{H}$ be a Hilbert space. Take a vector $y \in \mathcal{H}$. Define the functional

$$\varphi_y : \mathcal{H} \to \mathbb{K}, \quad \text{by} \quad \varphi_y(x) := \langle y|x\rangle, \quad \text{for} \quad x \in \mathcal{H}. \tag{1.3}$$

The Riesz representation theorem for Hilbert spaces states that all elements of $\mathcal{H}'$ are of the form φ_y in Eq. (1.3). The theorem identifies each functional $\varphi \in \mathcal{H}'$ with a unique map $\langle y|\cdot\rangle : \mathcal{H} \to \mathbb{K}$ that sends each vector $x \in \mathcal{H}$ to the inner product $\langle y|x\rangle$.

Theorem 1.3 (Riesz representation theorem)**.** *Let $\mathcal{H}$ be a Hilbert space and let $\varphi \in \mathcal{H}'$. There exists a unique $y \in \mathcal{H}$ such that $\varphi(x) = \langle y|x\rangle$ for all $x \in \mathcal{H}$. The map*

$$\mathcal{H} \to \mathcal{H}', \quad y \mapsto \varphi_y,$$

is an anti-linear isometric isomorphism between $\mathcal{H}$ and $\mathcal{H}'$.

Proof. By Reed and Simon (1980, Theorem II.4). □

1.2.4 Operators on Hilbert spaces

An operator in quantum mechanics maps one quantum state to another. For example, the Hamiltonian in the Schrödinger equation is an operator that guides the evolution of the quantum state. Other examples of operators are orthogonal projections in the measurement process in quantum mechanics.

Definition 1.8. Let $\mathcal{H}$ be a complex Hilbert space. A densely defined linear operator on $\mathcal{H}$ is a function $A : D(A) \to \mathcal{H}$ such that $D(A)$ is a dense vector subspace of $\mathcal{H}$ and for all $v, w \in D(A)$ and $\alpha, \beta \in \mathbb{C}$,

$$A(\alpha v + \beta w) = \alpha Av + \beta Aw.$$

Most of the operators in quantum mechanics are unbounded. Unbounded operators are not defined on the entirety of the relevant Hilbert space, but only on a dense subspace thereof. For example, operators representing position, momentum, and energy are unbounded operators.

Definition 1.9. A linear operator $A : \mathcal{H} \to \mathcal{H}$ on Hilbert space $\mathcal{H}$ is called bounded if

$$\sup_{\psi \in \mathcal{H} \setminus \{0\}} \frac{\|A\psi\|_{\mathcal{H}}}{\|\psi\|_{\mathcal{H}}} < \infty.$$

An operator that is not necessarily bounded is called unbounded. If the operator A is bounded, then the supremum defines the operator norm

$$\|A\| := \sup_{\psi \in \mathcal{H} \setminus \{0\}} \frac{\|A\psi\|_{\mathcal{H}}}{\|\psi\|_{\mathcal{H}}}.$$

The space of all bounded linear operators on Hilbert space $\mathcal{H}$ is denoted by $\mathcal{B}(\mathcal{H})$.

An observable in quantum mechanics provides a property or a quantity of the quantum state. For example, momentum, position, energy, or spin are observables. The second axiom in the quantum mechanical formalism states that any observable is a self-adjoint operator acting on a Hilbert space $\mathcal{H}$.

Axiom 2 (Postulate II)**.** *The set of observables $\mathcal{A}$ of a quantum mechanical system with Hilbert space $\mathcal{H}$ consists of all self-adjoint operators on $\mathcal{H}$.*

The self-adjointness guarantees that the spectrum of an operator is real. The axiom thus ensures that the measurements are real because the outcomes of a measurement of a pure quantum mechanical state are elements of the spectrum.

Definition 1.10. Let $A : D(A) \to \mathcal{H}$ be a densely defined linear operator on some Hilbert space $\mathcal{H}$. The adjoint $A^* : D(A^*) \to \mathcal{H}$ of A is the operator defined by

$$D(A^*) = \{x \in \mathcal{H} : \exists y \in \mathcal{H} \text{ with } \langle x|Az\rangle = \langle y|z\rangle \text{ for all } z \in D(A)\},$$

where

$$A^* : x \mapsto y.$$

The adjoint A^* is well-defined because the Riesz representation theorem uniquely determines $y = A^*x$ in Definition 1.10. The adjoint is also linear.

Definition 1.11. A densely defined operator $A : D(A) \to \mathcal{H}$ on a Hilbert space $\mathcal{H}$ is self-adjoint if

$$A = A^*.$$

Note that the definition of self-adjointness requires $D(A) = D(A^*)$.

1.2.5 Dirac bra-ket notation

Dirac (1939) introduced the bra-ket notation to streamline quantum mechanics. Recall that a vector in a Hilbert space $\mathcal{H}$ describes a physical (pure) state of a quantum mechanical system. Physicists denote such a vector as $|\psi\rangle \in \mathcal{H}$ and call the vector a ket. A functional in the dual space $\langle\varphi| \in \mathcal{H}'$ is called a bra. The Riesz representation theorem yields that the bra $\langle\varphi|$ and ket $|\psi\rangle$ together give the inner product

$$\langle\varphi|\,(|\psi\rangle) = \langle\varphi|\psi\rangle. \tag{1.4}$$

Example 3. *Let $\mathcal{H} = \mathbb{C}^d$ be a qudit. The Dirac notation for qudits manifests as*

$$|\psi\rangle = \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_d \end{pmatrix}, \qquad \langle\psi| = \begin{pmatrix} \overline{\psi}_1 & \dots & \overline{\psi}_d \end{pmatrix},$$

$$\langle\phi|\psi\rangle = \sum_{i=1}^{d} \overline{\phi}_i\, \psi_i, \qquad |\psi\rangle\langle\phi| = \begin{pmatrix} \psi_1\,\overline{\phi}_1 & \dots & \psi_1\,\overline{\phi}_d \\ \vdots & \ddots & \vdots \\ \psi_d\,\overline{\phi}_1 & \dots & \psi_d\,\overline{\phi}_d \end{pmatrix}.$$

The entries $\psi_i, \phi_i \in \mathbb{C}$ for $i = 1, ..., d$, and the bar in $\overline{\phi}_i$ denotes the complex conjugacy. Notice that the bra is a complex conjugate transpose of the ket, i.e.,

$$\langle\psi| = \left(\overline{|\psi\rangle}\right)^T. \tag{1.5}$$

The standard orthonormal basis for a qudit is $\{|x\rangle : x \in \{0, 1, \dots, d-1\}\}$, i.e.,

$$|0\rangle := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad |1\rangle := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots \quad , |d-1\rangle := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

A (pure) state $|\phi\rangle \in \mathcal{H} = \mathbb{C}^d$ can be expressed in the standard orthonormal basis as

$$|\psi\rangle \;=\; \sum_{i=0}^{d-1} \alpha_i\,|i\rangle \quad \textit{such that} \sum_{i=0}^{d-1} |\alpha_i|^2 = 1.$$

The constants $\alpha_i \in \mathbb{C}$ are called amplitudes. The state $|\psi\rangle$ is said to be in the superposition of basis states when at least two of the amplitudes α_i are nonzero.

In particular, a (pure) qubit state $|\phi\rangle \in \mathcal{H} = \mathbb{C}^2$ has a form

$$|\psi\rangle \;=\; \alpha|0\rangle + \beta|1\rangle \quad \textit{such that } |\alpha|^2 + |\beta|^2 = 1 \tag{1.6}$$

for some complex amplitudes α and β. The standard basis measurement (Walter and Ozols 2024, Equation (1.29)) of the pure qubit state (1.6) gives the Born's rule formulation

$$\mathbb{P}[x|\psi] := |\langle x|\psi\rangle|^2. \tag{1.7}$$

The measurement of the qubit gives only two possible measurement outcomes: $|0\rangle$ or $|1\rangle$. The state $|0\rangle$ represent bit 0 and the state $|1\rangle$ represents bit 1 in quantum information theory. The Born's rule (1.7) implies that measuring the superposed qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ yields measured state $|0\rangle$ with probability $|\alpha|^2$ and state $|1\rangle$ with probability $|\beta|^2$.

The qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ describes any object with spin-$\frac{1}{2}$. A standard physical example of a qubit is an electron, where $|0\rangle$ corresponds to spin up, and $|1\rangle$ corresponds to spin down along, for example, the z-axis.

A linear operator $A : D(A) \to \mathcal{H}$ maps a ket $|\psi\rangle$ to another ket $A\,|\psi\rangle$. Extend the action of A to bras by precomposing $\langle\varphi|$ with A as

$$A(\langle\varphi|) : \mathcal{H} \to \mathbb{K} \quad \text{defined by} \quad A(\langle\varphi|)(|\psi\rangle) \mapsto (\langle\varphi|)(A\,|\psi\rangle).$$

Notice that $(\langle\varphi|)(A\,|\psi\rangle) = \langle|\varphi\rangle|\,(A\,|\psi\rangle)\rangle =: \langle\varphi|\,A\,|\psi\rangle$ is an inner product in simplified Dirac notation as in Eq. (1.4). When $\mathcal{H}$ is finite-dimensional and the linear operator A is identified with a matrix, $\langle\varphi|\,A\,|\psi\rangle$ is a matrix multiplication. For example, qudits simplify to matrix multiplication.

Let $\psi, \varphi \in \mathcal{H}$. Then $|\psi\rangle\langle\varphi| : \mathcal{H} \to \mathcal{H}$ denotes a linear operator on Hilbert space $\mathcal{H}$ given by

$$(|\psi\rangle\langle\varphi|)(|\phi\rangle) = |\psi\rangle\langle\varphi|\phi\rangle = \langle\varphi|\phi\rangle|\psi\rangle,$$

for all $|\phi\rangle \in \mathcal{H}$.

Hall (2013, Notation 3.27-29) provides more notational remarks.

1.2.6 Spectral decomposition

Recall that self-adjoint operators describe quantum mechanical observables by Postulate II of quantum mechanics. The spectrum of a self-adjoint operator must be real because the spectrum of an observable corresponds to possible measurement outcomes.

Definition 1.12. Let $A : D(A) \to \mathcal{H}$ be a densely defined operator on some Hilbert space $\mathcal{H}$. The set

$$\rho(A) := \{\lambda \in \mathbb{K} : \lambda\,\mathrm{Id} - A \text{ is invertible}\}$$

is called the resolvent set of A. Its complement

$$\sigma(A) := \mathbb{K} \setminus \rho(A)$$

is called the spectrum of A.

The spectrum of an operator generalizes eigenvalues in the finite-dimensional case. Suppose $Av = \lambda v$ defines an eigenpair for $v \neq 0$. The operator $(\lambda \mathrm{Id} - A)v = \lambda v - Av = 0$, but also $(\lambda \mathrm{Id} - A)0 = 0$. Hence, $\lambda \mathrm{Id} - A$ is not injective. The lack of injectivity implies the nonexistence of the inverse of $\lambda \mathrm{Id} - A$. Therefore, $\lambda \notin \rho(A)$ and $\lambda \in \sigma(A)$. Every eigenvalue of A is in the spectrum $\sigma(A)$ of A. The spectrum $\sigma(A)$ of A consists exactly of the eigenvalues of A when $\mathcal{H}$ is finite dimensional.

Definition 1.13. A densely defined operator $A : D(A) \to \mathcal{H}$ on a Hilbert space $\mathcal{H}$ is called symmetric if

$$\langle x|Ay\rangle = \langle Ax|y\rangle \qquad \text{for all } x, y \in D(A).$$

Theorem 1.4. *Let $A : D(A) \to \mathcal{H}$ be a densely defined symmetric operator on some Hilbert space. Then*

$$\sigma(A) \subset \mathbb{R} \quad \textit{if and only if} \quad A \textit{ is self-adjoint.}$$

Proof. By Reed and Simon (1975, Theorem X.1(3)). □

Let $V \subseteq \mathcal{H}$ be a closed subspace. A unique bounded linear operator $P : \mathcal{H} \to \mathcal{H}$ exists that equals the identity on V and zero on $V^\perp$. The operator P is called the orthogonal projection onto V and satisfies $P^2 = P$ and $P^* = P$.

Definition 1.14 (Hall (2013, Definition 7.10))**.** Let $\mathcal{H}$ be a Hilbert space and (S, Σ) be a measurable space. A projection-valued measure (PVM) on S is a mapping $P : \Sigma \to \mathcal{B}(\mathcal{H})$, where $\mathcal{B}(\mathcal{H})$ denotes the set of bounded operators on $\mathcal{H}$, satisfying

1. the operator $P(\Omega)$ is an orthogonal projection for every $\Omega \in \Sigma$;
2. $P(\varnothing) = 0$ and $P(S) = I$;
3. for $(\Omega_n)_{n\in\mathbb{N}} \subset \Sigma$ pairwise disjoint,
$$P\left(\bigcup_{n\in\mathbb{N}} \Omega_n\right)\psi = \sum_{n=1}^{\infty} P(\Omega_n)\,\psi, \qquad \text{for all } \psi \in \mathcal{H};$$
4. and for all $\Omega_1, \Omega_2 \in \Sigma$,
$$P(\Omega_1)P(\Omega_2) = P(\Omega_1 \cap \Omega_2).$$

A PVM P assigns to every measurable set Ω an orthogonal projection $P(\Omega)$ and satisfies measure-like axioms. Any projection-valued measure $P : \Sigma \to \mathcal{B}(\mathcal{H})$ and a state $\psi \in \mathcal{H}$ form a nonnegative real-valued measure

$$P_\psi(\Omega) := \langle \psi|P(\Omega)\psi\rangle,$$

for all $\Omega \in \Sigma$.

The Spectral theorem associates each densely defined self-adjoint linear operator A with a unique projection-valued measure P^A.

Theorem 1.5 (Spectral Theorem)**.** *Suppose $A : D(A) \to \mathcal{H}$ is a self-adjoint linear operator on a Hilbert space $\mathcal{H}$. Then there is a unique projection-valued measure P^A: $\sigma(A) \to \mathcal{B}(\mathcal{H})$ such that*

$$A = \int_{\sigma(A)} \lambda \, dP^A(\lambda). \tag{1.8}$$

Proof. By Hall (2013, Theorem 10.4, Section 10.4). □

The operator A in Equation (1.8) is a result of operator-valued integration with respect to the projection-valued measure P^A. The next theorem defines the operator-valued integration for unbounded operators.

Theorem 1.6. *Suppose $P : \Sigma \to \mathcal{B}(\mathcal{H})$ is a projection-valued measure on a measurable space (S, Σ) with values in the set of all bounded operators $\mathcal{B}(\mathcal{H})$ on a Hilbert space $\mathcal{H}$. Let $f : S \to \mathbb{C}$ be a measurable function. Define a subspace W_f of $\mathcal{H}$ by*

$$W_f := \left\{ \psi \in \mathcal{H} \;\middle|\; \int_S |f(\lambda)|^2 \, dP_\psi(\lambda) < \infty \right\},$$

where $P_\psi(\Omega) := \langle \psi | P(\Omega) \psi \rangle$ for all $\Omega \in \Sigma$ is a scalar measure. Then there exists a unique unbounded operator $\int_S f \, dP : W_f \to \mathcal{H}$ with the property that

$$\left\langle \psi \middle| \left(\int_S f \, dP \right) \psi \right\rangle = \int_S f(\lambda) \, dP_\psi(\lambda),$$

for all $\psi \in W_f$.

Proof. By Hall (2013, Proposition 10.1, Proposition 10.2). □

The domain W_f is a dense subspace of a Hilbert space $\mathcal{H}$. If f is bounded, then $\int_S f \, \mathrm{d}P$ is a bounded operator with the domain $W_f = \mathcal{H}$. The operator satisfies the following properties (Hall 2013, Proposition 7.11).

1. For all $\Omega \in \Sigma$,
$$\int_S \mathbf{1}_\Omega \, \mathrm{d}P = P(\Omega).$$
In particular, the integral of the constant function 1 is the identity operator I on Hilbert space $\mathcal{H}$.

2. For all bounded and measurable $f : S \to \mathbb{C}$,
$$\left\| \int_S f \, \mathrm{d}P \right\| \leqslant \sup_{\lambda \in S} |f(\lambda)|.$$

These two properties imply the expected behavior of the integration on the simple and bounded measurable functions. For a simple function

$$s = \sum_{k=1}^{m} a_k \mathbb{1}_{A_k}, \qquad A_k \in \Sigma \text{ pairwise disjoint}, \; a_k \in \mathbb{C},$$

the integral

$$\int_S s \, \mathrm{d}P \; = \; \sum_{k=1}^{m} a_k \, P(A_k).$$

Functional calculus is a theory of applying mathematical functions to mathematical operators. The unique projection-valued measure in the spectral theorem is crucial for the definition of the functional calculus of operators in quantum mechanics.

Definition 1.15. If an operator $A : D(A) \to \mathcal{H}$ is self-adjoint and $f : \sigma(A) \to \mathbb{C}$ is a measurable function, define a possibly unbounded operator $f(A)$ by

$$f(A) := \int_{\sigma(A)} f(\lambda) \, \mathrm{d}P^A(\lambda),$$

where P^A is the projection-valued measure in the Spectral theorem 1.5.

1.2.7 Quantum states

Definition 1.16. A linear operator $A : \mathcal{H}_1 \to \mathcal{H}_2$ between two Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ is compact if $A(B)$ is a compact subset of $\mathcal{H}_2$, where B denotes the closed unit ball in $\mathcal{H}_1$. The space $\mathcal{K}(\mathcal{H}_1, \mathcal{H}_2)$ consists of all compact operators between $\mathcal{H}_1$ and $\mathcal{H}_2$.

The following definition supposes that the operator A is compact. Any compact operator is bounded. If A is bounded, then A^*A is self-adjoint. Hence, the functional calculus in Definition 1.15 defines $\sqrt{A^*A}$.

Definition 1.17. Let $\mathcal{H}$ be a separable, infinite-dimensional Hilbert space. A compact operator $A \in \mathcal{K}(\mathcal{H})$ is called trace class if

$$\|A\|_1 := \sum_{n=1}^{\infty} \left\langle (A^*A)^{1/2} e_n \middle| e_n \right\rangle < \infty$$

for an orthonormal basis $(e_n)_{n \in \mathbb{N}}$ of $\mathcal{H}$. The series

$$\operatorname{Tr} A := \sum_{n=1}^{\infty} \langle A e_n | e_n \rangle$$

is independent of the choice of orthonormal basis and is called the trace of A.

Pure quantum states are equivalently represented by unit vectors $|\psi\rangle \in \mathcal{H}$ or operators $|\psi\rangle\langle\psi|$ on $\mathcal{H}$. Postulate III gives rise to the so-called mixed states that are complementary to the pure states. Mixed states are all the positive trace-class operators ρ on $\mathcal{H}$ with $\operatorname{Tr}\rho = 1$ that cannot be expressed as $\rho = |\psi\rangle\langle\psi|$ for some unit vector $|\psi\rangle \in \mathcal{H}$. A trace-class operator ρ is positive if the operator ρ is self-adjoint and all eigenvalues of ρ are non-negative. The pure and mixed states together describe all the quantum states of the quantum system associated with the Hilbert space $\mathcal{H}$.

Axiom 3 (Postulate III)**.** *The set of states $\mathcal{S}$ of a quantum system with a Hilbert space $\mathcal{H}$ consists of all positive trace-class operators ρ with $\operatorname{Tr}\rho = 1$. Pure states are projection operators onto one-dimensional subspaces of $\mathcal{H}$. All other states are called mixed states.*

Pauli matrices are linear operators on the set of states $\mathcal{S}$ of the qubit quantum system associated with the Hilbert space $\mathbb{C}^2$.

Definition 1.18 ((Walter and Ozols 2024, Equation (1.15)))**.** Let S denote the set of states of the qubit quantum system associated with the Hilbert space $\mathcal{H} = \mathbb{C}^2$. The Pauli matrices X, Y, and Z are three linear operators on the qubit system $\mathcal{H}$ defined by

$$X := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y := \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad Z := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{1.9}$$

where i is the imaginary unit.

The Pauli matrices X, Y, and Z together with the identity operator

$$I := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

form a basis for the set of states $\mathcal{S}$ on the qubit quantum system.

The Pauli matrix $Z : \mathbb{C}^2 \to \mathbb{C}^2$ plays an important role in the subsequent sections of this thesis. The following proposition shows how the Pauli Z operator maps the standard orthonormal basis $\{|0\rangle, |1\rangle\}$ for the qubit system.

Proposition 1.7. *Let $\mathcal{H} = \mathbb{C}^2$ be the Hilbert space associated with the qubit system. The Pauli operator $Z : \mathcal{H} \to \mathcal{H}$ satisfies*

$$Z|x\rangle = (-1)^x |x\rangle \tag{1.10}$$

for $x \in \{0, 1\}$.

Proof. Recall the the standard basis for $\mathcal{H} = \mathbb{C}^2$ is

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The Pauli Z operator is given by the matrix

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Pauli Z operator acts on the basis as

$$Z\left|0\right\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left|0\right\rangle = (-1)^0 \left|0\right\rangle,$$

$$Z\left|1\right\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -\left|1\right\rangle = (-1)^1 \left|1\right\rangle.$$

Hence,

$$Z\left|x\right\rangle = (-1)^x \left|x\right\rangle$$

for $x \in \{0, 1\}$. □

Lemma 1.8 ((Walter and Ozols 2024, Lemma 1.13))**.** *Let $\mathcal{S}$ be the set of states of a qubit system with an associated Hilbert space $\mathcal{H} = \mathbb{C}^2$. Any state $\rho \in \mathcal{S}$ in the set of states of the qubit system takes the form*

$$\rho = \frac{1}{2}\big(I + r_X X + r_Y Y + r_Z Z\big),$$

where X, Y and Z are Pauli matrices and

$$r = \begin{pmatrix} r_X \\ r_Y \\ r_Z \end{pmatrix}$$

is an arbitrary vector of norm $\|r\| \leqslant 1$. Moreover, ρ is pure if and only if $\|r\| = 1$.

Proof. By Preskill (2015, Section 2.3.2 Bloch sphere).

Let

$$M_2(\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{C} \right\}$$

be the vector space of all 2×2 complex matrices.

The Pauli matrices I, X, Y and Z are linearly independent because

$$\alpha I + \beta X + \gamma Y + \delta Z = 0, \tag{1.11}$$

for $\alpha, \beta, \gamma, \delta \in \mathbb{C}$, has only the trivial solution. In particular, Equation (1.11) gives that

$$\begin{pmatrix} \alpha + \delta & \beta - i\gamma \\ \beta + i\gamma & \alpha - \delta \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

which implies that

$$\alpha + \delta = 0, \qquad \alpha - \delta = 0, \qquad \beta - i\gamma = 0, \qquad \beta + i\gamma = 0.$$

Notice that $\alpha = \delta = 0$ follows from $\alpha + \delta = 0$ and $\alpha - \delta = 0$, and $\beta = \gamma = 0$ follows from $\beta - i\gamma = 0$ and $\beta + i\gamma = 0$.

The set of 4 linearly independent Pauli matrices $\{I, X, Y, Z\}$ forms a basis for $M_2(\mathbb{C})$ because $M_2(\mathbb{C})$ has dimension $2 \cdot 2 = 4$. Hence, any state $\rho \in \mathcal{S}$ has a basis expansion

$$\rho = a_1 I + a_2 X + a_3 Y + a_4 Z, \tag{1.12}$$

for some $a_1, a_2, a_3, a_4 \in \mathbb{C}$.

Notice that the Pauli matrices X, Y and Z are traceless. Hence,

$$\mathrm{Tr}(\rho) = \mathrm{Tr}(a_1 I) = a_1 \mathrm{Tr}(I) = 2a_1.$$

Postulate III states that $\mathrm{Tr}(\rho) = 1$, which implies that $a_1 = \frac{1}{2}$. Rewrite the basis expansion (1.12) as

$$\rho = \frac{1}{2}(I + r_X X + r_Y Y + r_Z Z) \tag{1.13}$$

for $r_X, r_Y, r_Z \in \mathbb{R}$. The Spectral Theorem implies that the coefficients r_X, r_Y, r_Z are real because the operator ρ is self-adjoint.

Equation (1.13) becomes

$$\rho = \begin{pmatrix} 1 + r_Z & r_X - \mathrm{i} r_Y \\ r_X + \mathrm{i} r_Y & 1 - r_Z \end{pmatrix},$$

and the determinant

$$\begin{aligned} \det \rho &= \frac{1}{4}(1 - r_X^2 - r_Y^2 - r_Z^2) \\ &= \frac{1}{4}(1 - r \cdot r), \end{aligned} \tag{1.14}$$

where

$$r = \begin{pmatrix} r_X \\ r_Y \\ r_Z \end{pmatrix}.$$

Postulate III claims that $\rho \in \mathcal{S}$ is positive which means that all eigenvalues of ρ are nonnegative. The determinant $\det \rho \geqslant 0$ because a determinant of a matrix is a product of all the eigenvalues of the matrix. The nonnegative determinant implies that $\det \rho = \frac{1}{4}(1 - r \cdot r) \geqslant 0$. Solving the inequality yields $1 \geqslant r \cdot r \geqslant \sqrt{r \cdot r} = \|r\|$, as desired.

Suppose $\|r\| = 1$. Identity (1.14) gives that the determinant $\det \rho = 0$. Use the following identity (Pop and Furdui 2017, Lemma 2.7)

$$\det(A + xB) = \det A + \big(\mathrm{Tr}(A)\mathrm{Tr}(B) - \mathrm{Tr}(AB)\big)x + (\det B)\, x^2,$$

for $A, B \in M_2(\mathbb{C})$ and $x \in \mathbb{C}$, to expand the characteristic polynomial

$$\begin{aligned} p_\rho(t) &:= \det(tI - \rho) \\ &= t^2 + \det(\rho) + \mathrm{Tr}(tI)\mathrm{Tr}(-\rho) - \mathrm{Tr}(-tI\rho) \\ &= t^2 - 2t + \mathrm{Tr}(t\rho) \\ &= t^2 - 2t + t \\ &= t(t-1). \end{aligned}$$

The roots of the characteristic polynomial $p_\rho(t)$ are the eigenvalues 0 and 1 of ρ. The Spectral Theorem (Walter and Ozols 2024, Theorem 1.3) for finite-dimensional spaces provides a spectral decomposition

$$\rho = 1 \cdot |\psi\rangle \langle\psi| ,$$

for some state $|\psi\rangle \in \mathcal{H}$. Hence, ρ is a pure state.

Suppose ρ is pure. Then $\rho = |\psi\rangle \langle\psi|$ for some state $|\psi\rangle \in \mathcal{H}$. Hence, ρ has eigenvalues 0 and 1 by the Spectral Theorem (Walter and Ozols 2024, Theorem 1.3). The Invertible Matrix Theorem (Lay, Lay, and McDonald 2016, p. 277) implies that the $\det \rho = 0$ because ρ has eigenvalue 0. Identity (1.14) gives that $0 = \frac{1}{4}(1 - r \cdot r)$ and $1 = r \cdot r = \sqrt{r \cdot r} = \|r\|$ follows.

□

1.2.8 Composite systems

A quantum system is composite when the system consists of quantum subsystems. Postulate Ib formalizes composite systems with the tensor product of Hilbert spaces.

Axiom 4 (Postulate Ib)**.** *The Hilbert space of a composite system is the tensor product of Hilbert spaces of the single component systems.*

Example 4. *An n-qubit system consists of n quantum subsystems. Each subsystem is a qubit and is associated with the Hilbert space $\mathbb{C}^2$. Postulate Ib implies that the composite n-qubit system is associated with a Hilbert space $\mathcal{H} = \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 = (\mathbb{C}^2)^{\otimes n}$ which is a tensor product of n qubit Hilbert spaces.*

Recall that the standard orthonormal basis for a qubit (1.6) is $\{|0\rangle, |1\rangle\}$. The composite space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ has a natural product orthonormal basis

$$\{|x\rangle = |x_1, \ldots, x_n\rangle = |x_1\rangle \otimes \cdots \otimes |x_n\rangle : x = (x_1, \ldots, x_n) \in \{0,1\}^n\}.$$

Walter and Ozols (2024, Equation (2.3)) define the tensor product $A_1 \otimes A_2$ of linear operators $A_1 : \mathcal{H}_1 \to \mathcal{H}_1$ and $A_2 : \mathcal{H}_2 \to \mathcal{H}_2$ on finite-dimensional Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ by

$$(A_1 \otimes A_2)(|\psi_1\rangle \otimes |\psi_2\rangle) := A_1|\psi_1\rangle \otimes A_2|\psi_2\rangle \tag{1.15}$$

for all $|\psi_1\rangle \in \mathcal{H}_1$ and for all $|\psi_2\rangle \in \mathcal{H}_2$.

1.2.9 Measurement

Let $\mathcal{A}$ denote the set of observables and $\mathcal{S}$ denote the set of all quantum states. The following postulate describes the measurement of quantum systems. Recall that any observable A is self-adjoint and the spectral theorem associates each self-adjoint operator to a unique projection-valued measure P^A.

Axiom 5 (Postulate IV)**.** *The process of measurement in quantum mechanics is described by the assignment*

$$\mathcal{A} \times \mathcal{S} \ni (A, \rho) \mapsto \mu_A,$$

where μ_A *is the probability measure on* $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ *given by the Born–von Neumann formula*

$$\mu_A(\Omega) = \mathrm{Tr}\big(P_A(\Omega)\rho\big) \qquad \text{for } \Omega \in \mathscr{B}(\mathbb{R}),$$

where P_A *is the projection-valued measure associated to the self-adjoint operator* A*. For every Borel set* $\Omega \subseteq \mathbb{R}$*, the quantity* $0 \leqslant \mu_A(\Omega) \leqslant 1$ *is the probability that for a quantum system in the state* ρ *the result of a measurement of the observable* A *belongs to* Ω*.*

The trace is well-defined because $P_A(\Omega)\rho$ is trace class as the composition of a bounded and a trace class operator.

1.2.10 The Schrödinger equation

The Schrödinger equation

$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t} |\psi(t)\rangle = H(t)|\psi(t)\rangle, \tag{1.16}$$

where $\hbar = \frac{h}{2\pi}$ is the Planck's constant h divided by 2π, describes the time evolution of the quantum system. The evolution of the quantum system is deterministic and determined by a self-adjoint operator H on $D(\mathcal{H})$ called the Hamiltonian of the system.

2 Quantum annealing

Many problems in finance, operations research, and machine learning reduce to combinatorial optimization over discrete variables. For binary optimization with n variables, the solution space contains 2^n configurations. Exhaustive enumeration becomes computationally intractable even for moderate n. For example, a problem with $n = 60$ variables has approximately 10^{18} configurations.

Classical optimization algorithms execute discrete steps to explore minima. The quantum annealing algorithms encode the optimization problem onto a ground state of a quantum system. The system evolves continuously according to the Schrödinger equation (1.16). In particular, the quantum system starts in an easily prepared ground quantum state and gradually transforms into a ground state encoding the solution to the optimization problem. The measurement of the system at the end of the evolution yields the minima.

This chapter introduces quantum annealing as a quantum computational paradigm for solving combinatorial optimization problems. Section 2.1 describes the theoretical foundation via the quantum adiabatic theorem, and Section 2.2 provides the quantum annealing algorithm.

2.1 The quantum adiabatic theorem

The quantum adiabatic theorem provides the conditions for the theoretical guarantee that quantum annealing finds optimal solutions. This section develops the necessary mathematical framework.

2.1.1 Time-dependent Hamiltonians and instantaneous eigenstates

Consider a quantum system with Hilbert space $\mathcal{H}$ evolving by a time-dependent Hamiltonian operator $H(t) : D(H(t)) \to \mathcal{H}$ for $t \in [0, T_f]$, where $T_f > 0$ is the total evolution time. The observable $H(t)$ is self-adjoint for each t by Postulate II (Section 2). Assume the domain $D(H(t))$ is independent of time. Recall that the quantum state $|\psi(t)\rangle \in \mathcal{H}$ evolves according to the time-dependent Schrödinger equation (1.16)

$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

The overall Hilbert space $\mathcal{H}$ is $2^n < \infty$ dimensional in quantum annealing because the quantum annealing system consists of $n \in \mathbb{N}$ qubit $\mathbb{C}^2$ subsystems. The overall Hilbert space is given by the tensor product $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ by quantum mechanical axioms. The rest of this chapter assumes that the Hilbert space $\mathcal{H}$ is finite dimensional.

Since $H(t)$ is self-adjoint for each fixed t, the spectral theorem (1.5) guarantees that $H(t)$ possesses a complete spectral decomposition. The spectral decomposition of the Hamiltonian $H(t)$ simplifies in a finite-dimensional Hilbert space.

Theorem 2.1. *Let $H(t) : \mathcal{H} \to \mathcal{H}$ be a Hamiltonian operator on an 2^n-dimensional Hilbert space $\mathcal{H}$. The Hamiltonian has a spectral decomposition*

$$H(t) = \sum_{j=1}^{2^n} E_j(t)|\phi_j(t)\rangle\langle\phi_j(t)| \tag{2.1}$$

for real eigenvalues $E_1(t), ..., E_{2^n}(t)$ and a corresponding orthonormal eigenvector basis $|\phi_1(t)\rangle, ..., |\phi_{2^n}(t)\rangle$.

Proof. By Walter and Ozols (2024, Thereom 1.3). □

The spectral decomposition (2.1) is also called eigendecomposition because the instantaneous eigenpairs $(E_j(t), |\phi_j(t)\rangle)$ satisfy the eigenvalue equation.

Definition 2.1. For each time $t \in [0, T_f]$, an instantaneous eigenstate $|\phi_j(t)\rangle \in \mathcal{H}$ satisfies the eigenvalue equation

$$H(t)|\phi_j(t)\rangle = E_j(t)|\phi_j(t)\rangle,$$

where $E_j(t) \in \mathbb{R}$ is the corresponding instantaneous energy eigenvalue.

Order the eigenvalues such that $E_1(t) \leqslant E_2(t) \leqslant \cdots \leqslant E_{2^n}(t)$. The instantaneous ground state is $|\phi_1(t)\rangle$ with the corresponding lowest possible eigenvalue energy $E_1(t)$. The eigenvalues $E_1(t), ..., E_{2^n}(t)$ are called energies because the eigenvalues are the only possible outcomes of the energy measurement of the system. Any eigenstate $|\phi_j(t)\rangle$ of the system that has a higher corresponding energy $E_j(t) > E_1(t)$ than the ground state is called an excited state.

Definition 2.2. An instantaneous eigenstate $|\phi_j(t)\rangle \in \mathcal{H}$ is called degenerate if there exists a distinct instantaneous eigenstate $|\phi_k(t)\rangle \in \mathcal{H}$ such that the corresponding instantaneous energy eigenvalues are the same, $E_j(t) = E_k(t)$, for some time $t \in [0, T_f]$.

The eigenstates that are not degenerate are called non-degenerate. Plot 2.1 shows an example of energy eigenvalues evolution. The non-degeneracy of the ground eigenstate is a crucial assumption in the adiabatic theorem.

2.1.2 Introduction to adiabaticity

Adiabatic evolution, adiabatic approximation, and the adiabatic theorem are distinct concepts.

A quantum system initially in the ground eigenstate $|\phi_1(0)\rangle$ of a time-dependent Hamiltonian $H(t)$ undergoes adiabatic evolution if a "sufficiently slow" variation of the

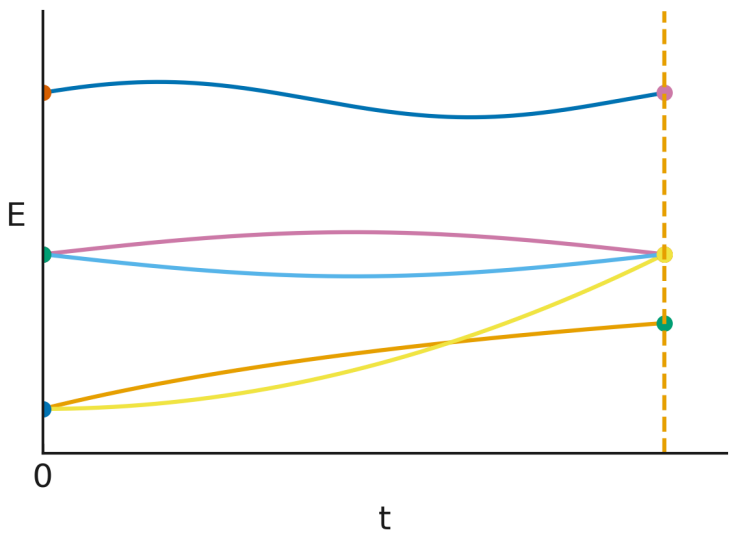

Figure 2.1: The plot shows energy eigenvalues $E_j(t)$ for $j \in \{1, ..., 5\}$ as lines evolving in time $t \in [0, T_f]$. The ground eigenstate $|\phi_1(t)\rangle$ is degenerate because its corresponding energy eigenvalue $E_1(t)$ equals the energy eigenvalue $E_2(t)$ of the first excited eigenstate $|\phi_2(t)\rangle$ at some time t in the evolution. The plot depicts the degeneracy as the intersection of the yellow and orange lines. The excited ground eigenstate $|\phi_5(t)\rangle$ is nondegenerate because the eigenstate does not "share" its energy eigenvalue $E_5(t)$ with any other eigenstate during the evolution. The plot depicts nondegeneracy as the absence of line intersections.

Hamiltonian $H(t)$ approximately keeps the actual state of the system $|\psi(t)\rangle$ in the instantaneous ground eigenstate $|\phi_1(t)\rangle$ of $H(t)$ during the evolution governed by the Schrödinger equation (1.16).

The adiabatic approximation provides a sufficient condition for the adiabatic evolution.

The adiabatic theorem roughly states that the quantum system remains in the state evolving from the initial state $|\psi_j(0)\rangle$ if the Hamiltonian $H(t)$ evolves "infinitely slowly".

Definition 2.3. Let $t \in [0, T_f]$ be time, and $E_1(t) \leqslant E_2(t) \leqslant ... \leqslant E_n(t)$ be instantaneous energy eigenvalues of the Hamiltonian $H(t)$ on an n-dimensional Hilbert space $\mathcal{H}$. Assume that the ground state $|\phi_1(t)\rangle$ is nondegenerate. The instantaneous spectral gap at time t is

$$\Delta(t) := E_2(t) - E_1(t).$$

The spectral gap quantifies the energy difference between the ground state and the first excited state when the ground state is nondegenerate and the Hilbert space is finite dimensional. A larger spectral gap provides a stronger separation between these states. The Hamiltonian $H(t)$ needs to evolve more slowly for adiabaticity when the spectral

gap $\Delta(t)$ is smaller.

2.1.3 Differentiation of operators on finite dimensional Hilbert spaces

Recall that the Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ in quantum annealing is finite-dimensional. The finite-dimensionality of the Hilbert space yields the boundedness of linear operators on the Hilbert space.

Theorem 2.2. *If a Hilbert space $\mathcal{H}$ is finite-dimensional, then every linear operator $A : \mathcal{H} \to \mathcal{H}$ on the Hilbert space $\mathcal{H}$ is bounded.*

Proof. By Kreyszig (1978, Theorem 2.7-8). □

The boundedness of the Hamiltonian operator $H(t)$ for every time t follows from Theorem 2.2 because the Hamiltonian $H(t)$ in quantum annealing is a linear operator on the finite-dimensional Hilbert space $\mathcal{H}$. The finite dimensional Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ has a finite orthonormal basis $\{e_i\}_{i=1}^{2^n}$. The Hamiltonian $H(t)$ on a finite-dimensional Hilbert space $\mathcal{H}$ can be represented by a matrix. The matrix representation of the Hamiltonian $H(t)$ depends on the choice of the orthonormal basis $\{e_i\}_{i=1}^{2^n}$ for the Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$.

Lemma 2.3. *Let $\mathcal{H}$ be an n-dimensional Hilbert space. Suppose $\{e_1, e_2, \ldots, e_n\}$ is an orthonormal basis for $\mathcal{H}$, and $A : \mathcal{H} \to \mathcal{H}$ is a bounded linear operator.*

Since $Ae_j \in \mathcal{H}$ for each $j \in \{1, \ldots, n\}$, there are unique scalars $m_{ij} \in \mathbb{C}$, for $i \in \{1, 2, \ldots, n\}$ and $j \in \{1, 2, \ldots, n\}$, such that

$$\begin{aligned} Ae_1 &= m_{11}e_1 + m_{21}e_2 + \cdots + m_{n1}e_n, \\ Ae_2 &= m_{12}e_1 + m_{22}e_2 + \cdots + m_{n2}e_n, \\ &\vdots \\ Ae_n &= m_{1n}e_1 + m_{2n}e_2 + \cdots + m_{nn}e_n. \end{aligned}$$

Let M be the $(n \times n)$ matrix whose j-th column consists of the complex scalars $m_{1j}, m_{2j}, \ldots, m_{nj}$ in the expansion of Ae_j, i.e.,

$$M = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1n} \\ m_{21} & m_{22} & \cdots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & \cdots & m_{nn} \end{bmatrix}.$$

Then

$$A|\psi\rangle = M|\psi\rangle \qquad \text{for all } |\psi\rangle \in \mathcal{H}. \tag{2.2}$$

Moreover, matrix M is the unique matrix satisfying relationship (2.2).

Proof. By Frazier (1999, Lemma 1.49). □

Lemma 2.3 implies that the Hamiltonian $H(t) : \mathcal{H} \to \mathcal{H}$ on the finite-dimensional Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ is a $(2^n \times 2^n)$ matrix with complex entries. The derivative

$$\dot{H}(t) := \frac{\mathrm{d}H(t)}{\mathrm{d}t} = \begin{bmatrix} \frac{\mathrm{d}h_{1,1}(t)}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}h_{1,2^n}(t)}{\mathrm{d}t} \\ \vdots & \ddots & \vdots \\ \frac{\mathrm{d}h_{2^n,1}(t)}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}h_{2^n,2^n}(t)}{\mathrm{d}t} \end{bmatrix}$$

is the usual matrix-by-scalar derivative, also known as the tangent matrix, where

$$H(t) = \begin{bmatrix} h_{1,1}(t) & \cdots & h_{1,2^n}(t) \\ \vdots & \ddots & \vdots \\ h_{2^n,1}(t) & \cdots & h_{2^n,2^n}(t) \end{bmatrix}$$

is the matrix representation of the Hamiltonian $H(t)$.

2.1.4 Hamiltonian interpolation

The initial Hamiltonian $H_I : \mathcal{H} \to \mathcal{H}$ is a self-adjoint operator whose ground state is simple to prepare. The problem Hamiltonian $H_P : \mathcal{H} \to \mathcal{H}$ is a self-adjoint operator whose ground state encodes the solution to the optimization problem. Finding the ground state of H_P directly is intractable. Quantum annealing finds the ground state using the adiabatic evolution.

The time-dependent annealing Hamiltonian

$$H(t) = \left(1 - \frac{t}{T_f}\right) H_I + \frac{t}{T_f} H_P, \quad t \in [0, T_f], \tag{2.3}$$

interpolates between the initial Hamiltonian H_I and the problem Hamiltonian H_P.

Define the dimensionless annealing parameter $s := t/T_f \in [0, 1]$. The annealing Hamiltonian

$$H(s) = (1 - s)H_I + sH_P.$$

The system is governed entirely by H_I at $s = 0$ and by H_P at $s = 1$. The evolution smoothly transitions between these extremes.

More general annealing schedules use smooth monotonic functions $A(s)$ and $B(s)$ satisfying $A(0) = 1$, $A(1) = 0$, $B(0) = 0$, $B(1) = 1$ to evolve the annealing Hamiltonian

$$H(s) = A(s)H_I + B(s)H_P. \tag{2.4}$$

2.1.5 Adiabatic theorem

The adiabatic theorem exists in rigorous or approximate variants. For example, Jansen, Ruskai, and Seiler (2007) provides a rigorous formulation of the adiabatic theorem. The theorem relates the spectral gap $\Delta(t)$ and the total evolution time T_f to the closeness of the system's actual state to the system's ground state.

Nondimensionalization of the physical units yields the Schrödinger equation (1.16) in natural units

$$\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H(t)|\psi(t)\rangle \tag{2.5}$$

where the constant $\hbar = 1$. Natural units simplify equations by setting physical constants to 1. Set the annealing parameter $s := \frac{t}{T_f}$. The Schrödinger equation (2.5) becomes

$$\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}s}|\hat{\psi}(s)\rangle = T_f\hat{H}(s)|\hat{\psi}(s)\rangle$$

for dimensionless parameter $s \in [0,1]$ because $\mathrm{d}t = T_f\mathrm{d}s$, $|\hat{\psi}(s)\rangle = |\psi(T_f s)\rangle = |\psi(t)\rangle$, and $\hat{H}(s) = H(T_f s) = H(t)$.

Theorem 2.4. *Let $H(s)$ be a twice-differentiable Hamiltonian operator on a finite-dimensional Hilbert space $\mathcal{H}$ for all $s \in [0,1]$. Assume the instantaneous ground state of $H(s)$ is nondegenerate for all $s \in [0,1]$ and the instantaneous spectral gap $\Delta(s) > 0$ for all $s \in [0,1]$. Let $|\phi(s)\rangle$ denote the instantaneous ground state of $H(s)$ and take the initial condition $|\psi(0)\rangle = |\phi(0)\rangle$. If the total evolution time T_f (in units with $\hbar = 1$) satisfies*

$$T_f \geqslant \frac{2}{\varepsilon}\left[c_1\frac{\|\dot{H}(0)\|}{\Delta(0)^2} + c_1\frac{\|\dot{H}(1)\|}{\Delta(1)^2} + \int_0^1\left((3c_1^2 + c_1 + c_3)\frac{\|\dot{H}(s)\|^2}{\Delta(s)^3} + c_2\frac{\|\ddot{H}(s)\|}{\Delta(s)^2}\right)ds\right],$$

then the solution of the Schrödinger equation

$$\mathrm{i}\,\tfrac{d}{ds}|\psi(s)\rangle = T_f\,H(s)|\psi(s)\rangle$$

obeys

$$\big\|\,|\psi(1)\rangle - |\phi(1)\rangle\,\big\| \leqslant \varepsilon.$$

Here $\|\cdot\|$ denotes the operator norm, and the constants $c_1, c_2, c_3 > 0$ arise from first- and second-order perturbation bounds

$$\|\dot{P}\| \leqslant c_1\frac{\|\dot{H}\|}{\Delta} \qquad \|\ddot{P}\| \leqslant c_2\frac{\|\ddot{H}\|}{\Delta} + c_3\frac{\|\dot{H}\|^2}{\Delta^2},$$

where $P(s) := |\phi(s)\rangle\langle\phi(s)|$ is the ground-state projection.

Proof. By Childs (2008). □

Check the lecture notes by Childs (2008) for details on the constants $c_1, ..., c_3$ in the perturbation bounds on the derivatives of the ground-state projection $P(s) := |\phi(s)\rangle\langle\phi(s)|$.

2.2 Optimization with quantum annealing

Quantum annealing applies the adiabatic theorem to solve optimization problems. The algorithm encodes the optimization objective as the ground state energy of a problem Hamiltonian and uses adiabatic evolution to find that ground state.

2.2.1 Quantum annealer

Quantum computing encompasses two computing paradigms: gate-based computing and quantum annealing. Gate-based quantum computing uses sequences of quantum logic gates to manipulate qubits and perform computations. IBM, Google, and others are pursuing the gate-based model. Quantum annealing is a computational method for solving optimization problems by finding ground states. D-Wave Quantum Inc. (2025a) is a quantum computing company that develops quantum annealing hardware and software. Aharonov et al. (2007, Theorem 1.1, Theorem 1.3) suggests that adiabatic quantum computation can physically realize universal quantum computation.

A quantum annealer is a quantum computing device that uses quantum annealing to solve quadratic unconstrained binary optimization (QUBO) problems (2.6). Figure 2.2 shows D-Wave's quantum annealer.

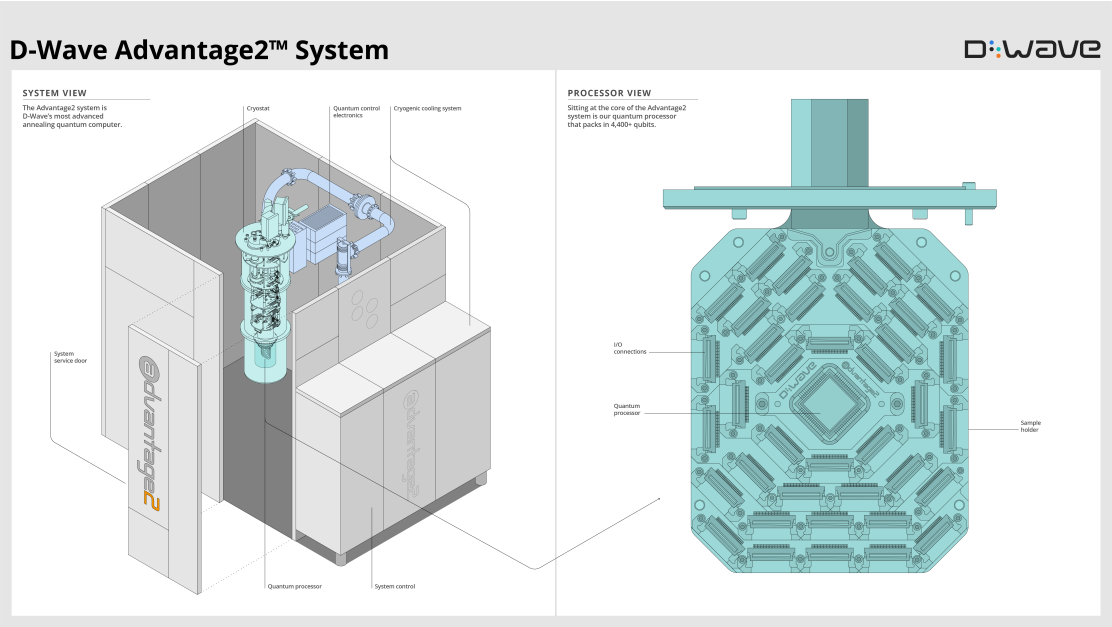

Figure 2.2: The Advantage2 system (D-Wave Quantum Inc. 2025b), available via the Leap quantum cloud service (D-Wave Quantum Inc. 2025c), is D-Wave's most advanced annealing quantum computer. The Advantage2 system features 4,400+ physical qubits.

2.2.2 Quadratic unconstrained binary optimization (QUBO)

The quantum annealers solve quadratic unconstrained binary optimization (QUBO) problems. The QUBO problem is NP-hard (Lewis and Glover 2017). The QUBO model is a mathematical formulation of optimization problems in which the objective function consists of quadratic terms, there are no explicit constraints, and the decision variables take binary values 0 and 1. The objective function often incorporates the constraints

of an optimization problem via penalty terms. Binary decision variables often discretize continuous decision variables of the optimization problem.

There are several ways to express the QUBO model.

Definition 2.4 (Glover, Kochenberger, and Du (2018))**.** The QUBO model is

$$\min_{x\in\{0,1\}^n} / \max \quad f(x) = x^T Q x \tag{2.6}$$

where $x \in \{0,1\}^n$ is the decision variable and Q is an $n \times n$ real coefficient matrix.

Other formulations are

$$\min_{x\in\{0,1\}^n} / \max \quad H = x^T q + x^T Q x,$$

where $q \in \mathbb{R}^n$, and a combination of these terms

$$\min_{x\in\{0,1\}^n} / \max \quad H = \lambda_1 \cdot H_1 + \lambda_2 \cdot H_2 + \cdots ,$$

where, for $i = 1, 2, ...$, weight $\lambda_i \in \mathbb{R}$, and $H_i = x^T q_i + x^T Q_i x$ for $q_i \in \mathbb{R}$ and Q_i an $n \times n$ real coefficient matrix. The weights tune the optimization problem and include constraints in the QUBO model.

Matrix Q is symmetric or upper triangular by convention. The asymmetric part of Q does not affect the QUBO formulation.

Proposition 2.5. *Suppose $x \in \mathbb{R}^n$ and Q is a real $n \times n$ matrix. Let $Q^{sym} := \frac{1}{2}(Q + Q^T)$ represent the symmetric part of Q and $Q^{asym} := Q - Q^{sym}$ represent the asymmetric part of Q. Then the asymmetric part does not affect the QUBO formulation (2.6), i.e., $x^T Q x = x^T Q^{sym} x$.*

Proof. Notice that $Q^{\text{asym}} = Q - \frac{1}{2}(Q + Q^T) = \frac{1}{2}(Q - Q^T)$. Hence, $Q^{\text{asym}^T} = -\frac{1}{2}(Q - Q^T) = -Q^{\text{asym}}$. Since $x^T Q^{\text{asym}} x$ is a scalar,

$$\begin{aligned} x^T Q^{\text{asym}} x &= (x^T Q^{\text{asym}} x)^T \\ &= x^T Q^{\text{asym}^T} x \\ &= -x^T Q^{\text{asym}} x. \end{aligned}$$

Addition of $x^T Q^{\text{asym}} x$ to both sides yields that $x^T Q^{\text{asym}} x = 0$. Therefore, $x^T Q x = x^T Q^{\text{sym}} x + x^T Q^{\text{asym}} x = x^T Q^{\text{sym}} x$. □

The following transformations achieve the symmetric and upper triangular forms without loss of generality. The transformation replacing q_{ij} by

$$\frac{q_{ij} + q_{ji}}{2} \tag{2.7}$$

for all i and j, except for $i = j$, yields symmetric Q. The transformation replacing q_{ij} by

$$q_{ij} + q_{ji} \tag{2.8}$$

for all i and j with $j > i$, and by 0 for $j < i$ yields upper triangular Q.

The symmetrization (2.7) and upper-triangularization (2.8) show that any QUBO model (2.6) is equivalent to a transformed QUBO model with an upper-triangular matrix Q. The upper-triangular form of the matrix Q is a convention in the QUBO model definition. The conventional QUBO model is

$$\min_{x \in \{0,1\}^n} \quad f(x) = x^T Q x = \sum_{i=1}^{n} Q_{ii} x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} x_i x_j \tag{2.9}$$

where $x \in \{0,1\}^n$ is the decision variable and Q is an upper triangular $n \times n$ real coefficient matrix. Notice that $x^2 = x$ for any $x \in \{0,1\}$.

Objective function $7x_1 - 9x_2 + 4x_3 + x_1x_2 + 2x_1x_3 - 1.5x_2x_3$ is an example of the QUBO model (2.6). Figure 2.3 shows a graphical representation of this QUBO example.

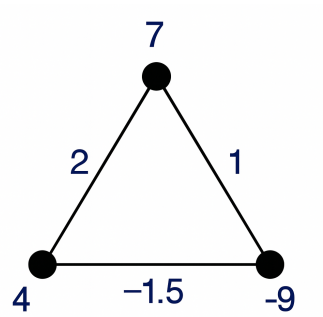

Figure 2.3: Image shows a graphical representation of the QUBO model $7x_1 - 9x_2 + 4x_3 + x_1x_2 + 2x_1x_3 - 1.5x_2x_3$ where coefficients of the quadratic terms appear on edges, and the coefficients of linear terms appear on nodes.

2.2.3 Initial Hamiltonian

Recall that the initial Hamiltonian $H_I : \mathcal{H} \to \mathcal{H}$ governs the quantum system at the beginning of quantum annealing.

Definition 2.5 (Morita and Nishimori (2008, Equation (19))). Suppose $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ is Hilbert space associated with an n-qubit system. The initial Hamiltonian $H_I : \mathcal{H} \to \mathcal{H}$ is given by

$$H_I = \sum_{i=1}^{n} X_i,$$

where operator $X_i := I^{\otimes(i-1)} \otimes X \otimes I^{\otimes(n-i)}$ denotes the Pauli X operator applied to the i-th qubit of the n-qubit system.

The adiabatic theorem requires that the quantum system is initialized in the ground state of the initial Hamiltonian H_I. Albash and Lidar (2018) states that the ground state of the initial Hamiltonian H_I is the uniform superposition state in the standard basis.

Definition 2.6 (Wolf (2012))**.** Suppose $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ is the Hilbert space associated with an n-qubit system. Recall that $\{|x\rangle : x \in \{0,1\}^n\}$ is the standard product orthonormal basis for $\mathcal{H}$. The uniform superposition state is

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle.$$

The uniformity means that each of the n qubits is equally likely to be measured in state $|0\rangle$ or state $|1\rangle$.

2.2.4 Construction of the problem Hamiltonian H_P

The conventional QUBO model (2.9) is

$$\min_{x \in \{0,1\}^n} \quad f(x) = x^T Q x = \sum_{i=1}^{n} Q_{ii} x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} x_i x_j,$$

where $x \in \{0,1\}^n$ is the decision variable and Q is an upper triangular $n \times n$ real coefficient matrix.

Quantum annealers construct a problem Hamiltonian operator H_P so that the lowest energy eigenvalue E_1 of H_P is equal to the objective function $f(x^*)$ in the the conventional QUBO model (2.9) evaluated at its minimizer $x^* \in \{0,1\}^n$.

Definition 2.7 ((Hadfield 2021, Equation (1)))**.** Let $f : \{0,1\}^n \to \mathbb{R}$ be a function acting on n bits. Problem Hamiltonian $H_P : \mathcal{H} \to \mathcal{H}$ on Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ represents function f if

$$H_P|x\rangle = f(x)|x\rangle$$

for each input string $x \in \{0,1\}^n$ with corresponding standard orthonormal basis state $|x\rangle$ for $\mathcal{H}$.

Definition 2.7 defines every eigenvalue of the problem Hamiltonian as a value of the QUBO objective function f (2.6). The next theorem explicitly constructs a problem Hamiltonian H_P representing the QUBO objective function f (2.6).

Theorem 2.6 ((Hadfield 2021, Corollary 5))**.** *Suppose $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ is Hilbert space associated with an n-qubit system. Let the QUBO model be*

$$\min_{x \in \{0,1\}^n} \quad f(x) = x^T Q x = \sum_{i=1}^{n} Q_{ii} x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} x_i x_j,$$

where $x \in \{0,1\}^n$ is the decision variable and Q is an upper triangular $n \times n$ real coefficient matrix. Operator $Z_i : \mathcal{H} \to \mathcal{H}$ is defined by

$$Z_i := I^{\otimes(i-1)} \otimes Z \otimes I^{\otimes(n-i)},$$

and denotes the Pauli Z operator applied to the i-th qubit of the n-qubit system. Then the problem Hamiltonian $H_P : \mathcal{H} \to \mathcal{H}$ representing the objective function f is

$$H_P = (q_{diag} + q_{off})I_{\mathcal{H}} - \frac{1}{2}\sum_{i=1}^{n}\left(Q_{ii} + \frac{1}{2}\sum_{j=i+1}^{n} Q_{ij} + \frac{1}{2}\sum_{j=1}^{i-1} Q_{ji}\right) Z_i + \frac{1}{4}\sum_{i=1}^{n}\sum_{j=i+1}^{n} Q_{ij} Z_i Z_j, \tag{2.10}$$

where $q_{diag} := \frac{1}{2}\sum_{i=1}^{n} Q_{ii}$ and $q_{off} := \frac{1}{4}\sum_{i=1}^{n}\sum_{j=i+1}^{n} Q_{ij}$.

The following proof mimics the proof by Hadfield (2021, Theorem 1 and Theorem 3). The differences in the proof by Hadfield arise from different QUBO forms. In particular, this thesis uses the upper-triangular matrix Q to gather the coefficients in the QUBO model, so the second coefficient in Equation (2.10) does not simplify because $Q_{ij} \neq Q_{ji}$.

2.2.5 Proof of Theorem 2.6

Let $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ be a Hilbert space associated with an n-qubit system. Operator $Z_i : \mathcal{H} \to \mathcal{H}$ is a tensor product of operators (1.15) defined by

$$Z_i := I^{\otimes(i-1)} \otimes Z \otimes I^{\otimes(n-i)}.$$

Operator Z_i applies the Pauli Z operator (1.9) to the i-th qubit of the n-qubit system.

Lemma 2.7 provides a Hamiltonian $H_{x_i} : \mathcal{H} \to \mathcal{H}$ representing function $f(x) = x_i$ where $x \in \{0,1\}^n$.

Lemma 2.7. *Let function $f : \{0,1\}^n \to \{0,1\}$ be defined by $f(x) = x_i$ for an $i \in \{1,\dots,n\}$. The problem Hamiltonian*

$$H_{x_i} = \frac{1}{2}I - \frac{1}{2}Z_i \tag{2.11}$$

represents function f.

Proof. The tensor product operator Z_i acts on the standard product basis of $\mathcal{H}$ as

$$\begin{aligned}
Z_i |x\rangle &= (I^{\otimes(i-1)} \otimes Z \otimes I^{\otimes(n-i)})(|x_1\rangle \otimes \cdots \otimes |x_i\rangle \otimes \cdots \otimes |x_n\rangle) \\
&= I^{\otimes(i-1)}(|x_1\rangle \otimes \cdots \otimes |x_{i-1}\rangle) \otimes Z |x_i\rangle \otimes I^{\otimes(n-1)}(|i+1\rangle \otimes \cdots \otimes |x_n\rangle) \\
&= |x_1\rangle \otimes \cdots \otimes |x_{i-1}\rangle \otimes (-1)_i^x |x_i\rangle \otimes |i+1\rangle \otimes \cdots \otimes |x_n\rangle \\
&= (-1)^{x_i} |x\rangle
\end{aligned}$$

for $x \in \{0,1\}^n$. The calculation above follows from the the definition (1.15) of the action of the tensor product operators and the property (1.10) of the action of the Pauli-Z operator on the standard basis for qubits.

Therefore,

$$H_{x_i} |x\rangle = \left(\frac{1}{2}I - \frac{1}{2}Z_i\right) |x\rangle = \frac{1}{2} |x\rangle - \frac{1}{2}(-1)^{x_i} |x\rangle = \frac{1-(-1)^{x_i}}{2} |x\rangle .$$

Notice that

$$\frac{1-(-1)^{x_i}}{2} = x_i$$

because

$$\frac{1-(-1)^0}{2} = \frac{1-1}{2} = 0$$
$$\frac{1-(-1)^1}{2} = \frac{1+1}{2} = 1.$$

Thus $H_{x_i} |x\rangle = x_i |x\rangle$ for all $x \in \{0,1\}^n$. □

Lemma 2.8 provides a Hamiltonian $H_{x_i x_j} : \mathcal{H} \to \mathcal{H}$ representing function $f(x) = x_i x_j$ where $x \in \{0,1\}^n$.

Lemma 2.8. *Let function $f : \{0,1\}^n \to \{0,1\}$ be defined by $f(x) = x_i x_j$ for $i, j \in \{1, \ldots, n\}$ with $i < j$. The problem Hamiltonian*

$$H_{x_i x_j} = \frac{1}{4}(I - Z_i - Z_j + Z_i Z_j)$$

represents function f.

Proof. Recall that the operators H_{x_i} and H_{x_j} are matrices in finite-dimensional Hilbert spaces. Matrices H_{x_i} and H_{x_j}, from Equation (2.11), are diagonal. Let operator $H : \mathcal{H} \to \mathcal{H}$ on a finite-dimensional Hilbert space $\mathcal{H}$ represent a function f, i.e., $H |x\rangle = f(x) |x\rangle$, for all basis elements $|x\rangle$. Then the matrix entry $H_{ji} = \langle y| H |x\rangle$ where j and i are the indexes of 1 in the standard basis vectors $|y\rangle$ and $|x\rangle$, respectively. The off-diagonal entries $H_{ji} = \langle y| H |x\rangle = \langle y| f(x) |x\rangle = f(x) \langle y|x\rangle = 0$ because $\langle y|x\rangle = 0$ when $x \neq y$.

Suppose operator H_f, H_g, and H_{fg} represent functions $f : \{0,1\}^n \to \mathbb{R}$, $g : \{0,1\}^n \to \mathbb{R}$, and $fg : \{0,1\}^n \to \mathbb{R}$, respectively. Operators $H_f H_g = H_{fg}$ because

$$H_f H_g |x\rangle = H_f(g(x) |x\rangle) = g(x) H_f |x\rangle = g(x) f(x) |x\rangle = (fg)(x) |x\rangle = H_{fg} |x\rangle .$$

When diagonal operators $H_f H_g$ and H_{fg} act the same on the standard basis, they are the same.

Application of the result $H_f H_g = H_{fg}$ to the function f in Lemma 2.8 yields

$$\begin{aligned} H_{x_i x_j} &= H_{x_i} H_{x_j} \\ &= \left(\frac{1}{2}I - \frac{1}{2}Z_i\right)\left(\frac{1}{2}I - \frac{1}{2}Z_j\right) \\ &= \frac{1}{4}(I - Z_i - Z_j + Z_i Z_j) . \end{aligned}$$

The following table verifies the result of the lemma. Recall that $Z_i |x\rangle = (-1)^{x_i} |x\rangle$. Thus $Z_i Z_j |x\rangle = (-1)^{x_i+x_j} |x\rangle$ and

$$H_{x_i x_j} |x\rangle = \frac{1}{4}\left(1 - (-1)^{x_i} - (-1)^{x_j} + (-1)^{x_i+x_j}\right) |x\rangle .$$

Define $\phi(x_i, x_j) := \frac{1}{4}\left(1 - (-1)^{x_i} - (-1)^{x_j} + (-1)^{x_i+x_j}\right)$. Then

x_i	x_j	$(-1)^{x_i}$	$(-1)^{x_j}$	$(-1)^{x_i+x_j}$	$\phi(x_i, x_j)$	$x_i x_j$
0	0	1	1	1	$\frac{1}{4}(1-1-1+1) = 0$	0
0	1	1	-1	-1	$\frac{1}{4}(1-1+1-1) = 0$	0
1	0	-1	1	-1	$\frac{1}{4}(1+1-1-1) = 0$	0
1	1	-1	-1	1	$\frac{1}{4}(1+1+1+1) = 1$	1

In all cases, $\phi(x_i, x_j) = x_i x_j$, as required. □

Lemma 2.9 states that the mapping $f \mapsto H_f$ of Hamiltonian representation is linear.

Lemma 2.9. *Suppose Hamiltonians $H_{f_1}, \ldots, H_{f_m}$ represent functions $f_1, \ldots, f_m : \{0,1\}^n \to \mathbb{R}$, respectively. Then the Hamiltonian*

$$H_f = \sum_{k=1}^{m} w_k H_{f_k}$$

represents function $f = \sum_{k=1}^{m} w_k f_k$ for any $w_1, \ldots, w_m \in \mathbb{R}$.

Proof. Take any $x \in \{0,1\}^n$. Then

$$\begin{aligned} H_f |x\rangle &= \left(\sum_{k=1}^{m} w_k H_{f_k}\right) |x\rangle \\ &= \sum_{k=1}^{m} w_k f_k(x) |x\rangle \\ &= \left(\sum_{k=1}^{m} w_k f_k(x)\right) |x\rangle \\ &= f(x) |x\rangle . \end{aligned}$$

Hence, $H_f = \sum_{k=1}^{m} w_k H_{f_k}$ represents function f. □

Proof of Theorem 2.6. The QUBO objective function is

$$f(x) = x^T Q x = \sum_{i=1}^{n} Q_{ii} x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} x_i x_j .$$

By Lemma 2.9,

$$H_f = \sum_{i=1}^{n} Q_{ii} H_{x_i} + \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} H_{x_i x_j} .$$

By Lemmas 2.7 and 2.8:

$$H_f = \sum_{i=1}^{n} Q_{ii} \left(\frac{1}{2} I - \frac{1}{2} Z_i \right) + \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} \cdot \frac{1}{4} \left(I - Z_i - Z_j + Z_i Z_j \right).$$

The full expansion is

$$\begin{aligned} H_f = & \frac{1}{2} \sum_{i=1}^{n} Q_{ii} I - \frac{1}{2} \sum_{i=1}^{n} Q_{ii} Z_i \\ & + \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} I - \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} Z_i \\ & - \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} Z_j + \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} Z_i Z_j. \end{aligned} \tag{2.12}$$

The subsequent parts collect coefficients of the expansion (2.12). The identity operator I has the coefficients

$$q_{\text{diag}} := \frac{1}{2} \sum_{i=1}^{n} Q_{ii}, \qquad q_{\text{off}} := \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij}.$$

The linear coefficients of Pauli-Z operators are

$$\begin{aligned} & -\frac{1}{2} \sum_{i=1}^{n} Q_{ii} Z_i - \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} Z_i - \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} Z_j \\ = & -\frac{1}{2} \sum_{i=1}^{n} Q_{ii} Z_i - \frac{1}{4} \sum_{i=1}^{n} \sum_{j=i+1}^{n} Q_{ij} Z_i - \frac{1}{4} \sum_{i=1}^{n} \sum_{j=1}^{i-1} Q_{ji} Z_i \end{aligned}$$

after reindexation.

Each quadratic Pauli $Z_i Z_j$ term has the coefficient

$$\frac{1}{4} Q_{ij}.$$

□

2.2.6 Quantum annealing procedure

The quantum annealing algorithm consists of three stages. Let $|\psi(t)\rangle \in \mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ denote the actual state of the $n-$qubit quantum system on the quantum processor.

The algorithm in the initialization stage, at $t = 0$, prepares the physical realization of the n-qubit quantum system in the ground state $|\psi(0)\rangle = |\phi_1^I\rangle$ of the initial Hamiltonian H_I.

The second stage is the adiabatic evolution for $0 \leqslant t \leqslant T_f$. The system evolves according to the time-dependent Schrödinger equation (2.5). For large enough annealing time T_f, the Adiabatic theorem 2.4 guarantees the actual state $|\psi(t)\rangle \approx |\phi_1(t)\rangle$ evolves close to the instantaneous ground state $|\phi_1(t)\rangle$ of the interpolated Hamiltonian $H(t)$ (2.4) throughout the evolution.

The final stage of quantum annealing at $t = T_f$ measures the states of qubits. The actual quantum state $|\psi(T_f)\rangle \approx |\phi_1(T_f)\rangle$ is expected to be close to the ground state $|\phi_1(T_f)\rangle$ of the problem Hamiltonian H_P given by Equation (2.10). The measurement of qubits in the standard basis yields a bitstring $x \in \{0,1\}^n$ which represents the collapsed classical states of the qubit systems. Definition 2.7 of the Hamiltonian representation implies that the lowest energy eigenvalue of the problem Hamiltonian H_P is the objective function $f(x^*)$ of the QUBO model (2.9) evaluated at its minimizer $x^* \in \{0,1\}^n$. The state measurement of the system x provides the quantum annealing estimate $f(x) \approx f(x^*)$ of the the minimum of the QUBO objective function.

Quantum annealing may yield an excited state instead of a ground state because of imperfect adiabaticity, noise, or decoherence. The quantum processor performs multiple runs of the quantum annealing procedure and computes the lowest energy.

3 Portfolio optimization

An investor seeks to generate a profit on their investment. They can trade on a market that consists of a myriad of possible investments. These investments can be anything from government bonds and stocks to art pieces and cryptocurrencies. A portfolio is any collection of these investments.

We can hope for or expect a return from each investment. For example, an investor buys Apple stock for €200, hoping to sell the same stock for more next year. The risk is that the price of the Apple stock may drop, and the investor would make a loss when selling the stock. Portfolio optimization is the art of building a portfolio with some desired characteristics - usually maximizing the return and minimizing the risks.

Portfolio optimization research usually only considers financial assets traded on stock exchanges to construct portfolios. Stock exchanges allow easy, cheap, and fast trading compared to trading assets such as real estate or luxury goods. Two groups of assets are traded on stock exchanges: primitive assets and derivatives. Primitive assets are stocks, bonds, currencies, etc. A derivative is an asset that derives its value from (primitive) assets. Examples of derivatives are options, futures, or swaps.

A portfolio can be static or dynamic. We do not trade the assets in the static portfolio after its inception, but we trade assets in the dynamic portfolio during its existence based on new market information.

3.1 Static portfolio

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. All random variables are defined on this space and are real-valued. We build the portfolio at time $t = 0$ and close the portfolio at time $t = 1$. We do not adjust the portfolio in between.

The market consists of $d + 1$ assets, numbered from 0 to d. The zeroth asset is non-risky. Assume interest rate $r > -1$. Let price π_0 be the price of the zeroth asset at time $t = 0$. Then the price of the zeroth asset at $t = 1$ is $S_0 = \pi_0(1 + r) > 0$. We make a convention that $\pi_0 = 1$.

The other d assets are risky. Their prices π_i at $t = 0$ are known deterministic nonnegative numbers, but their prices S_i at $t = 1$ are not known at $t = 0$ and are modeled as nonnegative random variables. We define the return of risky asset i as $R_i = S_i - \pi_i$.

Let $\xi_i \in \mathbb{R}$ be the quantity of asset i in the portfolio. We are short-selling asset i if ξ_i is negative. Short-selling means borrowing an asset and selling it on the market at time $t = 0$, to repurchase it later at $t = 1$ at a potentially lower price. Notice that we can trade non-integer quantities of assets; for example, we can buy or sell $\log 12$ units of an asset.

Let $\bar{\pi} = (\pi_0, \pi)$ denote the price vector of the assets at $t = 0$, where π denotes the price vector of the d risky assets. Similarly, $\bar{S} = (S_0, S)$ is the price vector at $t = 1$. The vector $\bar{\xi} = (\xi_0, \xi)$ of asset quantities is called the portfolio. The random variables $W_0 = \bar{\xi} \cdot \bar{\pi}$ and $W_1 = \bar{\xi} \cdot \bar{S}$ are the values of the portfolio at times $t = 0$ and $t = 1$, respectively. The dot here denotes the ordinary inner product.

The return of the portfolio is $W_1 - W_0 = \bar{\xi} \cdot \left(\bar{S} - \bar{\pi}\right)$. The loss of the portfolio is $L := W_0 - W_1$ and its distribution function is F_L. Let $\bar{R} = (r, R)$ denote the vector of returns of the assets, where R is the vector of returns of the d risky assets. Then the portfolio loss is $L = -\bar{\xi} \cdot \bar{R}$ and portfolio return is $\bar{\xi} \cdot \bar{R}$.

3.2 Risk measures

Financial risks are factors that can lead to a loss in the value of assets and the portfolio. Market risk arises from fluctuations in market prices, such as changes in stock prices, interest rates, exchange rates, or commodity prices. Credit risk results from a borrower's failure to meet contractual debt obligations, such as a loan default. Operational risk stems from failures in internal processes, people, or external events. Portfolio optimization takes into account these and other financial risks in a holistic approach.

Let $\mathcal{L}^0(\Omega, \mathcal{F}, \mathbb{P})$ denote the set of all random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ that are a.s. finite. Each loss L in a linear space $\mathcal{M} \subset \mathcal{L}^0(\Omega, \mathcal{F}, \mathbb{P})$ represents the loss of a portfolio over some time horizon. Assume that constant random variables belong to $\mathcal{M}$ and denote them by lowercase letters. A risk measure is any mapping $\rho : \mathcal{M} \to \mathbb{R}$. Amount $\rho(L)$ represents the (regulatory) capital needed to back a portfolio with loss L. An alternative way to define a risk measure is as a mapping from net final portfolio values to $\mathbb{R}$.

A naive portfolio construction strategy is choosing an asset i with the greatest expected return $\mathbb{E}[R_i]$. The problem is that such an asset may have great variance $\mathrm{Var}[R_i]$ of its return. This would expose the investor to a risk of a great loss. Markowitz (1952) addressed this problem by using the variance of the portfolio's return as the risk measure. In particular, Markowitz constructed a portfolio that maximizes the portfolio's return and minimizes the variance of the portfolio's return.

There are several fundamental problems (Palomar 2025) with the Markowitz model. The first problem with the Markowitz model is that the variance risk measure penalizes desirable high returns. Assume that the density $p(\bar{\xi} \cdot \bar{R})$ of the relative portfolio return exists. The second problem is that the portfolio's variance relatively poorly characterizes (heavy) tails of the density, where the greatest losses occur. The third problem is that the solutions of the Markowitz model lack diversification.

Markowitz (1959) recognized these limitations and proposed the usage of semi-variance instead of variance. Semi-variance is a downside risk measure. A downside risk measure gives the risk of the portfolio underperforming a desired reference level. The benefit of such measures is that they do not penalize the outperformance of portfolios. Other downside risk measures are value at risk (VaR) and conditional value at risk (CVaR).

3.3 Value at risk (VaR)

VaR is probably the most widely used risk measure (McNeil, Frey, and Embrechts 2015) in financial institutions. It plays a vital role in the Basel regulatory framework. We can define VaR on the distribution of relative/absolute returns, losses, or the final values of the portfolio.

Consider a portfolio over a fixed time horizon. The VaR of the portfolio with loss $L \in \mathcal{M}$ at confidence level $\alpha \in (0,1)$ is the smallest number l such that the probability that the loss L exceeds l is no larger than $1-\alpha$. Formally,

$$\mathrm{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} : \mathbb{P}(L > l) \leqslant 1-\alpha\} = \inf\{l \in \mathbb{R} : F_L(l) \geqslant \alpha\},$$

which is simply the generalized inverse distribution function $F_L^{\leftarrow}(\alpha)$ of the loss. Typical values (McNeil, Frey, and Embrechts 2015) for α are 0.95 or 0.99. In market-risk management, the time horizon is usually one or ten days, and in credit and operational risk management, the time horizon is usually one year. The drawback of VaR is that it does not give any information about the severity of losses that occur with a probability of less than $1-\alpha$.

For example, let $\mathrm{VaR}_{0.95} = €2.5$ for the time horizon 10 days. The interpretation is that we can expect a loss of at most €2.5 with probability 0.95 in the next 10 days. The loss will exceed €2.5 in the next 10 days with probability 0.05. Figure 3.1 shows this example. Beware that the interpretation neglects estimation error, model risk, and market liquidity risk.

3.4 Expected shortfall (ES)

Assume $\mathbb{E}(|L|) < \infty$ for a loss $L \in \mathcal{M}$ with distribution function F_L. The expected shortfall (ES) or conditional value at risk (CVaR) at confidence level $\alpha \in (0,1)$ is

$$\mathrm{ES}_\alpha(L) = \mathrm{CVaR}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 F_L^{\leftarrow}(u)\,\mathrm{d}u,$$

where $F_L^{\leftarrow}(u) = \inf\{x \in \mathbb{R} : F_L(x) \geqslant u\} = \mathrm{VaR}_u(L)$ is the generalized inverse distribution function.

The condition $\mathbb{E}(|L|) < \infty$ ensures that the integral is well-defined. Let $U \sim U(0,1)$ be a uniform random variable. The inverse transform sampling method (McNeil, Frey, and Embrechts 2015, Proposition 7.2) implies that

$$L \stackrel{d}{=} F_L^{\leftarrow}(U),$$

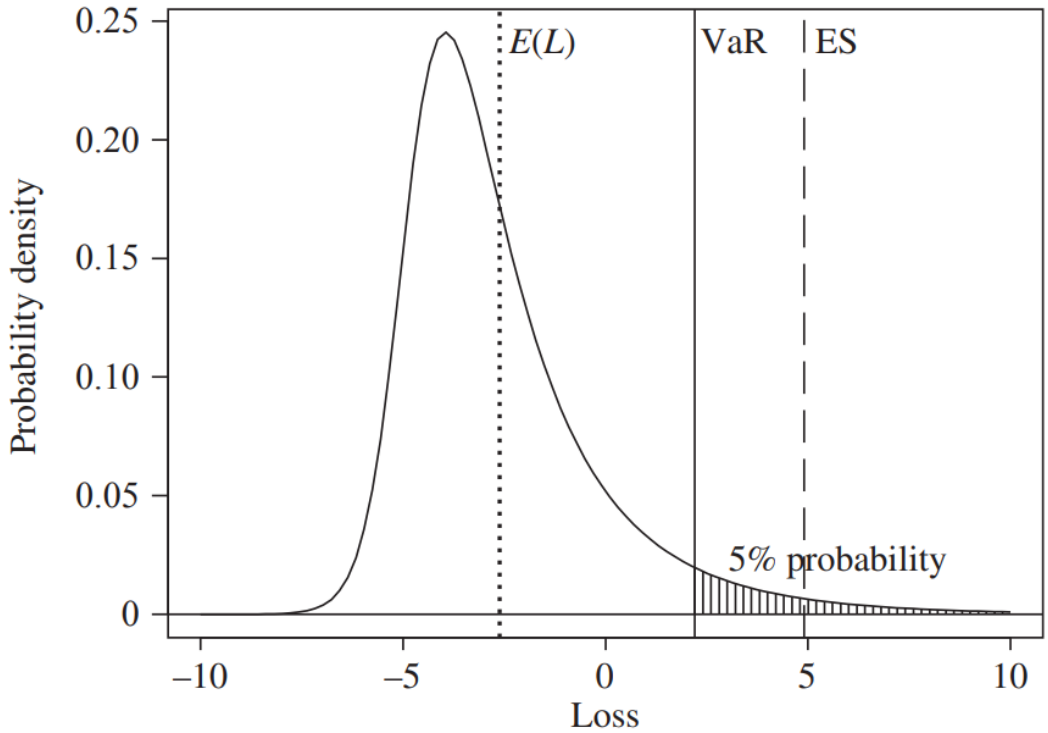

Figure 3.1: Assume that the density function $p(L)$ of the loss L of the portfolio exists. The plot (McNeil, Frey, and Embrechts 2015) shows density $p(L)$, expected value $\mathbb{E}[L]$ of the loss, value at risk $\mathrm{VaR}_{0.95}(L)$, and expected shortfall $\mathrm{ES}_{0.95}(L)$.

where $F_L^{\leftarrow}(U) = \inf\{x \in \mathbb{R} : F_L(x) \geqslant u\}$. Use the distribution equality to obtain

$$\begin{aligned}
\infty &> \mathbb{E}[|L|] \\
&= \mathbb{E}[|F_L^{\leftarrow}|] \\
&= \int_0^1 |F_L^{\leftarrow}(u)| \mathrm{d}u \\
&\geqslant \int_\alpha^1 |F_L^{\leftarrow}(u)| \mathrm{d}u,
\end{aligned}$$

for any $\alpha \in (0,1)$. The finiteness implies that $\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 F_L^{\leftarrow}(u)\,\mathrm{d}u < \infty$.

Notice that

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,\mathrm{d}u$$

because $F_L^{\leftarrow}(u) = \mathrm{VaR}_u(L)$. This means that ES_α averages VaR_u over all levels $u \geqslant \alpha$ and thus provides information about the tail of the loss distribution. An important remark is that $\mathrm{ES}_\alpha \geqslant \mathrm{VaR}_\alpha$, which follows from the fact that VaR_α is nondecreasing as

a quantile function. The fact that ES bounds VaR implies that minimizing ES minimizes the upper bound of VaR.

Assume that the loss distribution is continuous. We interpret ES as the expected loss when the loss exceeds VaR. For example, $\mathrm{ES}_{95\%}$ is the average loss in the 5% worst-case scenarios, as shown in Figure 3.1.

Lemma 3.1. *If loss L has a continuous distribution function F_L and $\mathbb{E}(|L|) < \infty$, then*

$$ES_\alpha(L) = \mathbb{E}[L \mid L \geqslant VaR_\alpha],$$

for any $\alpha \in (0, 1)$.

Proof. By McNeil, Frey, and Embrechts (2015, Proof of Lemma 2.13).

Let $U \sim U(0, 1)$ be a uniform random variable. The inverse transform sampling method (McNeil, Frey, and Embrechts 2015, Proposition 7.2) implies that

$$L \stackrel{d}{=} F_L^{\leftarrow}(U)$$

where $F_L^{\leftarrow}(U) = \inf\{x \in \mathbb{R} : F_L(x) \geqslant u\}$.

The continuity of F_L implies that $F_L^{\leftarrow}$ is strictly increasing (McNeil, Frey, and Embrechts 2015, Proposition A.3(iii)). Hence,

$$\{U \geqslant \alpha\} = \{F_L^{\leftarrow}(U) \geqslant F_L^{\leftarrow}(\alpha)\}.$$

Use these two facts to obtain

$$\begin{aligned}
\int_\alpha^1 \mathrm{VaR}_u(L)\mathrm{d}u &= \int_\alpha^1 F_L^{\leftarrow}(u)\mathrm{d}u \\
&= \int_0^1 F_L^{\leftarrow}(u)\mathbb{1}_{\alpha \leqslant u \leqslant 1}\mathrm{d}u \\
&= \mathbb{E}[F_L^{\leftarrow}(U)\mathbb{1}_{U \geqslant \alpha}] \\
&= \mathbb{E}[F_L^{\leftarrow}(U)\mathbb{1}_{F_L^{\leftarrow}(U) \geqslant F_L^{\leftarrow}(\alpha)}] \\
&= \mathbb{E}[L\mathbb{1}_{L \geqslant \mathrm{VaR}_\alpha(L)}] \\
&= \mathbb{E}[L|L \geqslant \mathrm{VaR}_\alpha(L)]\mathbb{P}[L \geqslant \mathrm{VaR}_\alpha(L)] \\
&= \mathbb{E}[L|L \geqslant \mathrm{VaR}_\alpha(L)](1 - \alpha).
\end{aligned}$$

□

We require the following two reformulations of ES for constructing the portfolio optimization problem.

Proposition 3.2. *Assume $\mathbb{E}(|L|) < \infty$ for a loss $L \in \mathcal{M}$ with distribution function F_L. For $0 < \alpha < 1$,*

$$ES_\alpha(L) = VaR_\alpha(L) + \frac{1}{1-\alpha}\mathbb{E}[(L - VaR_\alpha(L))^+],$$

where notation $(x)^+ := max(0, x)$.

Proof. By McNeil, Frey, and Embrechts (2015, Proposition 8.13).

Let $U \sim U(0,1)$. Recall that $F_L^{\leftarrow}(U) \stackrel{d}{=} L$. Hence,

$$\begin{aligned}\frac{1}{1-\alpha}\mathbb{E}[(L-\mathrm{VaR}_\alpha(L))^+] &= \frac{1}{1-\alpha}\int_0^1 (F_L^{\leftarrow}(u)-\mathrm{VaR}_\alpha(L))^+\mathrm{d}u \\ &= \frac{1}{1-\alpha}\int_\alpha^1 F_L^{\leftarrow}(u)-\mathrm{VaR}_\alpha(L)\mathrm{d}u \\ &= \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\mathrm{d}u-\mathrm{VaR}_\alpha(L).\end{aligned}$$

□

The following reformulation of ES makes VaR implicit as an optimal value of the decision variable τ. This reformulation is crucial for consequent piece-wise linear reformulation (3.2) and linear program (3.5).

Proposition 3.3. *Assume $\mathbb{E}(|L|) < \infty$ for a loss $L \in \mathcal{M}$ with distribution function F_L. For $0 < \alpha < 1$,*

$$ES_\alpha(L) = \min_{\tau\in\mathbb{R}}\left\{\tau + \frac{1}{1-\alpha}\mathbb{E}[(L-\tau)^+]\right\}, \tag{3.1}$$

where $VaR_\alpha(L) \in \underset{\tau\in\mathbb{R}}{argmin}\left\{\tau + \frac{1}{1-\alpha}\mathbb{E}[(L-\tau)^+]\right\}$.

Proof. Rockafellar and Uryasev (2000, Theorem 1).

This last paragraph develops an estimator of the ES reformulation (3.1) for the portfolio optimization model. Suppose $\{L_1, L_2, ..., L_T\}$ is a sample of i.i.d. realizations from the distribution μ_L of loss L. The strong law of large numbers (LLN) implies that

$$\frac{1}{T}\sum_{t=1}^{T}(L_t-\tau)^+ \xrightarrow{\text{a.s.}} \mathbb{E}[(L-\tau)^+],$$

as $T \to \infty$. We obtain a Monte-Carlo-based estimator

$$\hat{\mathrm{ES}}_\alpha(L) = \min_{\tau\in\mathbb{R}}\left\{\tau + \frac{1}{1-\alpha}\frac{1}{T}\sum_{t=1}^{T}(L_t-\tau)^+\right\}, \tag{3.2}$$

which is piece-wise linear in τ.

3.5 Coherent risk measures

Artzner et al. (1999) argues that a risk measure effectively manages risk if it is a so-called coherent measure. A risk measure ρ is coherent if it is monotone, translation invariant, subadditive, and positively homogeneous. Let $L, L_1, L_2 \in \mathcal{M}$ be losses, $m \in \mathbb{R}$, $\lambda \geqslant 0$, and $0 \leqslant \gamma \leqslant 1$.

A risk measure ρ is monotone if $L_1 \leqslant L_2$ a.s. implies that $\rho(L_1) \leqslant \rho(L_2)$. A monotone risk measure requires more backing capital from a portfolio with a higher loss. A risk measure ρ is translation invariant if $\rho(L+m) = \rho(L)+m$. Translation invariance enforces that adding capital m to a loss decreases the backing capital by the same amount. A risk measure ρ is subadditive if $\rho(L_1 + L_2) \leqslant \rho(L_1) + \rho(L_2)$. Subadditivity ensures that the merging of two portfolios does not increase risk and enforces the diversification principle. A risk measure ρ is positively homogeneous if $\rho(\lambda L) = \lambda\rho(L)$. Positive homogeneity ensures the risk scales linearly with the portfolio size.

A risk measure ρ is a convex risk measure on $\mathcal{M}$ if ρ is convex, monotone, and translation invariant on $\mathcal{M}$. A coherent risk measure is automatically convex (McNeil, Frey, and Embrechts 2015).

VaR is neither subadditive nor convex (Artzner et al. 1999), which means that VaR is not a coherent risk measure. Moreover, examples of VaR penalizing diversification contradict investment practice. Pflug (2000) proved that ES is a coherent risk measure. In this regard, ES is superior to VaR.

3.6 Mean-ES portfolio

Suppose the loss $L = -\bar{\xi} \cdot \bar{R}$ for some static portfolio optimization problem. The mean-ES bi-objective portfolio optimization maximizes the expected return and minimizes the expected shortfall. The optimization balances the expected loss of the top $1 - \alpha$ worst-case scenarios with the expected return. The optimization formulation is

$$\begin{aligned} \underset{\bar{\xi}\in\mathbb{R}^{d+1}}{\text{maximize}} \quad & \bar{\xi} \cdot \mathbb{E}[\bar{R}] - \lambda \mathrm{ES}_\alpha(-\bar{\xi} \cdot \bar{R}) \\ \text{subject to} \quad & \bar{\xi} \in \Xi \end{aligned}$$

where Ξ is a general set of constraints and $\lambda \in [0, \infty)$ is the trade-off parameter. A greater trade-off parameter λ decreases the expected shortfall at the expense of return, and vice versa.

Substitution of identity (3.1) yields equivalent portfolio optimization formulation

$$\begin{aligned} \underset{\bar{\xi}\in\mathbb{R}^{d+1},\, \tau\in\mathbb{R}}{\text{maximize}} \quad & \bar{\xi} \cdot \mathbb{E}[\bar{R}] - \lambda \left\{ \tau + \frac{1}{1-\alpha} \mathbb{E}[(-\bar{\xi} \cdot \bar{R} - \tau)^+] \right\} \\ \text{subject to} \quad & \bar{\xi} \in \Xi \end{aligned}$$

where an optimal τ is $\mathrm{VaR}_\alpha(L)$. Estimator (3.2) yields approximate portfolio optimization formulation

$$\begin{aligned} \underset{\bar{\xi}\in\mathbb{R}^{d+1},\, \tau\in\mathbb{R}}{\text{maximize}} \quad & \bar{\xi} \cdot \mathbb{E}[\bar{R}] - \lambda \left\{ \tau + \frac{1}{1-\alpha} \frac{1}{T} \sum_{t=1}^{T} (-\bar{\xi} \cdot \bar{r}_t - \tau)^+ \right\} \\ \text{subject to} \quad & \bar{\xi} \in \Xi \end{aligned} \tag{3.3}$$

where T is the sample size of i.i.d. observations $\{\bar{r}_1, ..., \bar{r}_T\}$ of assets return $\bar{R}$.

Let $x \in \mathbb{R}$. Introduce auxiliary variable $z \in \mathbb{R}$ to represent $\max(0, x)$ as

$$\begin{aligned} \underset{z \in \mathbb{R}}{\text{minimize}} \quad & z \\ \text{subject to} \quad & z \geqslant 0 \\ & z \geqslant x \end{aligned} \tag{3.4}$$

for any $x \in \mathbb{R}$. The optimal $z^* = \max(0, x)$.

Use the representation principle (3.4) to rewrite Problem (3.3) as

$$\begin{aligned} \underset{\bar{\xi} \in \mathbb{R}^{d+1},\, \tau \in \mathbb{R},\, y \in \mathbb{R}^T}{\text{maximize}} \quad & \bar{\xi} \cdot \mathbb{E}[\bar{R}] - \lambda \left\{ \tau + \frac{1}{1-\alpha} \frac{1}{T} \sum_{t=1}^{T} y_t \right\} \\ \text{subject to} \quad & y_t \geqslant 0 \\ & y_t \geqslant -\bar{\xi} \cdot \bar{r}_t - \tau \\ & \bar{\xi} \in \Xi \\ & \text{for } t = 1, ..., T. \end{aligned} \tag{3.5}$$

3.7 Budgetary constraint

Investors or trading desks have limited budgets. Problem (3.5) unrealistically allows trading arbitrary amounts of assets. We introduce a budgetary constraint $\bar{\xi} \cdot \bar{\pi} = B$ for some total budget $B > 0$. Problem (3.5) becomes

$$\begin{aligned} \underset{\bar{\xi} \in \mathbb{R}^{d+1},\, \tau \in \mathbb{R},\, y \in \mathbb{R}^T}{\text{maximize}} \quad & \bar{\xi} \cdot \mathbb{E}[\bar{R}] - \lambda \left\{ \tau + \frac{1}{1-\alpha} \frac{1}{T} \sum_{t=1}^{T} y_t \right\} \\ \text{subject to} \quad & y_t \geqslant 0 \\ & y_t \geqslant -\bar{\xi} \cdot \bar{r}_t - \tau \\ & \bar{\xi} \cdot \bar{\pi} = B \\ & \text{for } t = 1, ..., T. \end{aligned}$$

4 QUBO Mean-ES portfolio

4.1 Binary encoding

Binary encoding approximates a variable as a linear combination of binary variables. Suppose we want to binary encode variable $x \in [c, d]$ for $c, d \in \mathbb{R}$.

We can represent any integer $I \in \{0, 1, 2, ..., 2^K - 1\}$ with a binary vector $b = (b_0, ..., b_{K-1}) \in \{0, 1\}^K$ such that

$$I = \sum_{k=0}^{K-1} 2^k b_k.$$

Mapping

$$I \longmapsto c + \frac{d-c}{2^K - 1} I$$

from integers $\{0, 1, 2, ..., 2^K - 1\}$ onto a real range $[c, d]$ preserves the linearity of spacing.

Parameter $K \in \mathbb{N}$ is the number of binary variables used for the binary encoding. Greater values of K decrease the error of the approximation of variable x, but increase the number of qubits needed for the encoding. Choice of the parameter K is, hence, a trade-off parameter determined during the tuning of the model and is dependent on the qubits available on the hardware.

The approximate binary encoding (Carbone, De Grossi, and Spiller 2023, Eq. (16)) of $x \in [c, d]$ is

$$x^* = c + \frac{d-c}{2^K - 1} \sum_{k=0}^{K-1} 2^k b_k^*, \tag{4.1}$$

where binary vector $b^* = (b_0^*, ..., b_{K-1}^*) \in \{0, 1\}^K$ minimizes the distance between x and x^*. In particular, the minimizer b^* is an optimal value of the decision variable of the optimization problem

$$\underset{b=(b_0,...,b_{K-1})\in\{0,1\}^K}{\text{minimize}} \quad c + \frac{d-c}{2^K - 1} \sum_{k=0}^{K-1} 2^k b_k - x \tag{4.2}$$

for some fixed $K \in \mathbb{N}$.

The maximum absolute error

$$|x - x^*| \leqslant \frac{d-c}{2(2^K - 1)}$$

for an optimal binary encoding x^* because $\frac{d-c}{2^K-1}$ is the linear spacing of binary encoding (4.1). The error $|x - x^*| \to 0$ as $K \to \infty$. The error decreases for tighter bounds $[c, d]$ as well. The tighter bounds x has, the fewer K qubits we need to control the error of the binary encoding.

Choose parameter $K \geqslant \log_2(\frac{d-c}{2e} + 1)$ to bound the maximum absolute error by e, i.e.,

$$|x - x^*| \leqslant \frac{d-c}{2(2^K - 1)} \leqslant e.$$

4.2 Binary encoding decision variables

The Mean-ES portfolio optimization (3.5) contains decision variable $(\bar{\xi}, \tau, y) \in \mathbb{R}^{d+T+2}$. The QUBO model requires the decision variables to be binary. We can satisfy this requirement by binary encoding each decision variable. We derive and prove our bounds on these decision variables in this section. Binary encoding considerably increases the number of needed qubits.

Recall that $\{\bar{r}_1, \ldots, \bar{r}_T\}$ are T observations of the vector $\bar{R} = (r, R)$ of asset returns where vector R contains the returns of the d risky assets. Then $L = -\bar{\xi} \cdot \bar{R}$ is the portfolio loss, $\mathbb{E}[L] = -\bar{\xi} \cdot \mathbb{E}[\bar{R}]$ is the expected portfolio loss, and $\bar{\xi} \cdot \bar{r}_t$ is the t-th observation of the portfolio loss.

4.2.1 Binary portfolio

We constrain the portfolio $\xi \in \{0, 1\}^d$ of the risky assets. Variable $\xi_i = 1$ when the portfolio includes asset i, and $\xi_i = 0$ when the portfolio does not include asset i. This portfolio reflects situations when we do not trade arbitrary amounts of assets. This constraint saves scarce qubits by circumventing binary encoding of ξ. Binary encoding the portfolio ξ requires considerably more qubits than are currently available at the state-of-the-art quantum annealers.

Let the non-risky asset 0 be binary encoded. The budgetary constraint yields that $\xi_0 \in [B - \sum_{i=1}^d \pi_i, B]$. In this thesis's case study, $B = \sum_{i=1}^d \pi_i$. Hence, $\xi_0 \in [0, B]$. Mean-ES portfolio problem (4.2) becomes

$$\begin{aligned} \underset{\bar{\xi} \in [0,B] \times \{0,1\}^d,\, \tau \in \mathbb{R},\, y \in \mathbb{R}^T}{\text{maximize}} \quad & \bar{\xi} \cdot \mathbb{E}[\bar{R}] - \lambda \left\{ \tau + \frac{1}{1-\alpha} \frac{1}{T} \sum_{t=1}^{T} y_t \right\} \\ \text{subject to} \quad & y_t \geqslant 0 \\ & y_t \geqslant -\bar{\xi} \cdot \bar{r}_t - \tau \\ & \bar{\xi} \cdot \bar{\pi} = B \\ & \text{for } t = 1, ..., T. \end{aligned} \tag{4.3}$$

The binary-encoded $\xi_0 \in [0, B]$ takes form

$$\xi_0^* = \frac{B}{2^J - 1} \sum_{j=0}^{J-1} 2^j a_j, \tag{4.4}$$

where $a = (a_0, ..., a_{J-1}) \in \{0,1\}^J$ and the number of binary variables $J \in \mathbb{N}$ is a trade-off parameter for fine-tuning the optimization model.

4.2.2 Binary encoding τ

The following proposition says that VaR is at most the budget B.

Proposition 4.1. *Consider Problem (4.3). The optimal* $\tau = VaR_\alpha(-\bar{\xi} \cdot \bar{R}) \leqslant B$.

Proof. Notice that $L := -\bar{\xi} \cdot \bar{R} = -\bar{\xi} \cdot \bar{S} + \bar{\xi} \cdot \bar{\pi} \leqslant \bar{\xi} \cdot \bar{\pi} = B$. Hence, $F_L(B) = \mathbb{P}(L \leqslant B) = 1$. So,

$$\mathrm{VaR}_\alpha(L) := \inf\{x \in \mathbb{R} : F_L(x) \geqslant \alpha\} \leqslant B.$$

□

Binary encoding also requires a portfolio-independent lower bound of VaR.

Proposition 4.2. *Consider Problem (4.3). Let* λ_{max} *be the largest eigenvalue of the covariance matrix* $Cov(\bar{R})$. *The optimal*

$$\tau = VaR_\alpha(-\bar{\xi} \cdot \bar{R}) \geqslant C := -rB + \sum_{i=1}^{d} \mathbb{1}_{\{\mathbb{E}[R_i] > r\pi_i\}} r\pi_i - \mathbb{E}[R_i] - \sqrt{\frac{1-\alpha}{\alpha}} \sqrt{\lambda_{\max}(B^2 + d)}.$$

Proof. Let $0 < \alpha < 1$ be a confidence level of $\mathrm{VaR}_\alpha(L)$. Apply Cantelli's inequality to $Y := -L$ and obtain, for any $\lambda > 0$,

$$\mathbb{P}(Y - \mathbb{E}[Y] \geqslant \lambda) = \mathbb{P}(-L + \mathbb{E}[L] \geqslant \lambda) = \mathbb{P}(L - \mathbb{E}[L] \leqslant -\lambda) \leqslant \frac{\sigma_Y^2}{\sigma_Y^2 + \lambda^2} = \frac{\sigma_L^2}{\sigma_L^2 + \lambda^2},$$

where σ_Y^2 and σ_L^2 are variance of Y and L, respectively. Notice that the variances σ_Y^2 and σ_L^2 are the same.

Choose

$$\lambda_\varepsilon = \sigma_L \sqrt{\frac{1-\varepsilon\alpha}{\varepsilon\alpha}},$$

for some $0 < \varepsilon < 1$. Plugging in λ_ε yields

$$F_L(\mathbb{E}[L] - \lambda_\varepsilon) = \mathbb{P}(L - \mathbb{E}[L] \leqslant -\lambda_\varepsilon) \leqslant \frac{\sigma_L^2}{\sigma_L^2 + \sigma_L^2 \frac{1-\varepsilon\alpha}{\varepsilon\alpha}} = \varepsilon\alpha < \alpha,$$

because $0 < \varepsilon < 1$. This inequality implies that $\mathbb{E}[L] - \lambda_\varepsilon \notin \{x : F_L(x) \geqslant \alpha\}$. Notice $\mathbb{E}[L] - \lambda_\varepsilon \leqslant \mathrm{VaR}_\alpha(L) := \inf\{x : F_L(x) \geqslant \alpha\}$ because F_L is nondecreasing.

Taking a limit gives

$$\mathrm{VaR}_\alpha(L) \geqslant \lim_{\varepsilon \uparrow 1} \mathbb{E}[L] - \sigma_L \sqrt{\frac{1-\varepsilon\alpha}{\varepsilon\alpha}} = \mathbb{E}[L] - \sigma_L \sqrt{\frac{1-\alpha}{\alpha}}.$$

The portfolio-independent bound is

$$\mathrm{VaR}_\alpha(L) \geqslant \min_{\bar{\xi}\in[0,B]\times\{0,1\}^d} \mathbb{E}[-\bar{\xi}\cdot\bar{R}] - \sqrt{\frac{1-\alpha}{\alpha}} \max_{\bar{\xi}\in[0,B]\times\{0,1\}^d} \sqrt{\mathrm{Var}(-\bar{\xi}\cdot\bar{R})}. \tag{4.5}$$

The budgetary constraint $\bar{\xi}\cdot\bar{\pi} = B$ gives that $\xi_0 = B - \xi\cdot\pi$. Use this expression to rewrite

$$\begin{aligned}
\min_{\bar{\xi}\in[0,B]\times\{0,1\}^d} \mathbb{E}[-\bar{\xi}\cdot\bar{R}] &= \min_{\bar{\xi}\in[0,B]\times\{0,1\}^d} -\bar{\xi}\cdot\mathbb{E}[\bar{R}] \\
&= \min_{\bar{\xi}\in[0,B]\times\{0,1\}^d} -\xi_0 r - \xi\cdot\mathbb{E}[R] \\
&= \min_{\bar{\xi}\in[0,B]\times\{0,1\}^d} -(B-\xi\cdot\pi)r - \xi\cdot\mathbb{E}[R] \\
&= \min_{\bar{\xi}\in[0,B]\times\{0,1\}^d} -rB + \sum_{i=1}^d \xi_i(r\pi_i - \mathbb{E}[R_i]) \\
&= -rB + \sum_{i=1}^d \mathbb{1}_{\{\mathbb{E}[R_i]>r\pi_i\}} r\pi_i - \mathbb{E}[R_i].
\end{aligned} \tag{4.6}$$

This expression merely means that the portfolio with minimal expected loss is the portfolio that includes exactly the assets that outperform the non-risky bond in expectation.

Rewrite $\mathrm{Var}(-\bar{\xi}\cdot\bar{R}) = \bar{\xi}\cdot\mathrm{Cov}(\bar{R})\bar{\xi}$ in the matrix notation for the variance of a linear combination. Let $\lambda_{\max}$ be the greatest eigenvalue of $\mathrm{Cov}(\bar{R})$. Rayleigh (1896) showed that the Rayleigh quotient

$$\frac{\bar{\xi}\cdot\mathrm{Cov}(\bar{R})\bar{\xi}}{\bar{\xi}\cdot\bar{\xi}} \leqslant \lambda_{\max}$$

for any nonzero $\bar{\xi}$ and symmetric $\mathrm{Cov}(\bar{R})$. Reshuffling the terms gives $\mathrm{Var}(L) \leqslant \lambda_{\max}\bar{\xi}\cdot\bar{\xi}$. A straightforward upper bound is $\bar{\xi}\cdot\bar{\xi} = \xi_0^2 + \sum_{i=1}^d \xi_i^2 = \xi_0^2 + \sum_{i=1}^d \xi_i \leqslant B^2 + d$ because $\xi_i^2 = \xi_i$ for $\xi_i \in \{0,1\}$. Combining these two upper bounds yields $\mathrm{Var}(L) \leqslant \lambda_{\max}(B^2+d)$ and

$$\max_{\bar{\xi}\in[0,B]\times\{0,1\}^d} \sqrt{\mathrm{Var}(L)} \leqslant \sqrt{\lambda_{\max}(B^2+d)}. \tag{4.7}$$

Plug the variance (4.7) and expectation (4.6) bounds into the portfolio-independent bound (4.5) to obtain the proposition result. □

The binary-encoded $\tau \in [C, B]$ is

$$\tau^* = C + \frac{B-C}{2^K-1}\sum_{k=0}^{K-1} 2^k b_k \tag{4.8}$$

where $b = (b_0, ..., b_{K-1}) \in \{0,1\}^K$ is a minimizer of $|\tau - \tau^*(b)|$ and $K \in \mathbb{N}$ is the number of binary variables used in the binary encoding.

Notice that

$$K := \mathrm{Cov}(\bar{R}) = \begin{bmatrix} 0 & 0 \\ 0 & \Sigma \end{bmatrix}$$

where $\Sigma_{i,j} = \text{cov}[R_i, R_j]$ for $1 \leqslant i, j \leqslant d$ because $K_{1,1} = \text{cov}(r, r) = 0$, $K_{1,j+1} = \text{cov}(r, R_j) = r\mathbb{E}[R_j] - r\mathbb{E}[R_j] = 0$, and, by symmetry, $K_{j+1,1} = \text{cov}(R_j, r) = 0$ for any $1 \leqslant j \leqslant d$.

4.2.3 Binary encoding y_t

Equation (3.4) gives that the optimal $y_t = \max(0, -\bar{\xi} \cdot \bar{r}_t - \tau)$, which implies that the optimal $y_t \geqslant 0$. Recall from the proof of Proposition 4.1 that $L \leqslant B$ and from Proposition 4.2 that $\tau \geqslant C := -rB + \sum_{i=1}^{d} \mathbb{1}_{\{\mathbb{E}[R_i] > r\pi_i\}} r\pi_i - \mathbb{E}[R_i] - \sqrt{\frac{1-\alpha}{\alpha}} \sqrt{\lambda_{\max}(B^2 + d)}$. These inequalities bound

$$L - \tau \leqslant B - C. \tag{4.9}$$

Proposition 4.1 claims that an optimal $\tau = \text{VaR}_\alpha(L) \leqslant B$, and and Proposition 4.2 claims that an optimal $\tau = \text{VaR}_\alpha(L) \geqslant C$. Hence, $C \leqslant \tau \leqslant B$, and

$$0 \leqslant B - C. \tag{4.10}$$

Equations (4.9) and (4.10) imply that

$$\max(0, L - \tau) \leqslant B - C. \tag{4.11}$$

The binary-encoded $y_t \in [0, B - C]$ is

$$y_t^* = \frac{B - C}{2^L - 1} \sum_{l=0}^{L-1} 2^l c_{t,l}, \tag{4.12}$$

where vector $c_t = (c_{t,0}, ..., c_{t,L-1}) \in \{0, 1\}^L$ is a minimizer of $|y_t - y_t^*(c_t)|$, and $L \in \mathbb{N}$ is the number of binary variables used in the binary encoding.

4.3 Unconstraining the Mean-ES portfolio

We remove constraints from Problem (4.3) to obtain QUBO formulations. Recall that the constraints are $\bar{\xi} \cdot \bar{\pi} = B$ and $y_t \geqslant -\bar{\xi} \cdot \bar{r}_t - \tau$ for $t = 1, ..., T$. The aim is to reformulate the constraints as quadratic penalty terms and incorporate the terms into the objective function. The reformulation of the budgetary constraint is

$$P(\bar{\xi} \cdot \bar{\pi} - B)^2, \tag{4.13}$$

where $P \in \mathbb{R}_{\geqslant 0}$ is a large nonnegative penalty coefficient.

The reformulation of the inequality constraint yields two QUBO formulations. The first approach introduces slack variables into the penalty term. The disadvantage is that each slack variable requires extra scarce qubits. The second approach employs so-called unbalanced penalization (Montañez-Barrera et al. 2024), which requires no additional qubits. The disadvantage is that the unbalanced penalization slightly penalizes solutions that satisfy the constraint.

4.3.1 Slack-variables approach

Let $s_t \geqslant 0$ be a slack variable such that

$$P_t(y_t + \bar{\xi} \cdot \bar{r}_t + \tau - s_t)^2 \tag{4.14}$$

is a penalty term in QUBO with a large nonnegative penalty coefficient $P_t \in \mathbb{R}_{\geqslant 0}$. The slack variable makes the penalty term vanish when the constraint holds. In particular, when the constraint $y_t \geqslant -\bar{\xi} \cdot \bar{r}_t - \tau$ holds, the slack variable makes the penalty zero. When the constraint fails; $y_t + \bar{\xi} \cdot \bar{r}_t + \tau$ is negative, the slack variable evaluates to zero, and the penalty becomes positive.

The slack variable s_t takes values from zero to $y_t + \bar{\xi} \cdot \bar{r}_t + \tau$ to nullify the penalty term. The variable must be binary encoded for the QUBO model. The upper bound $y_t + \bar{\xi} \cdot \bar{r}_t + \tau$ must be independent of the decision variables for binary encoding. The following proposition gives such an upper bound.

Proposition 4.3. *Consider Problem (4.3) unconstrained with slack variables (4.14). The slack variable* $s_t \leqslant C_t := (2+r)B - C + \sum_{i=1}^{d} \mathbb{1}_{\{r_{t,i} > r\pi_i\}} r_{t,i} - r\pi_i$.

Proof. Equation (4.11) and Proposition (4.1) give $y_t \leqslant B - C$ and $\tau \leqslant B$, respectively. The budgetary constraint $\bar{\xi} \cdot \bar{\pi} = B$ implies that $\xi_0 = B - \xi \cdot \pi$. Substitute this identity into

$$\begin{aligned} \bar{\xi} \cdot \bar{r}_t &= \xi_0 r + \xi \cdot r_t = rB - \xi \cdot (r_t - r\pi) \\ &\leqslant rB + \sum_{i=1}^{d} \mathbb{1}_{\{r_{t,i} > r\pi_i\}} r_{t,i} - r\pi_i. \end{aligned}$$

Substitute these three upper bounds into $s_t \leqslant y_t + \bar{\xi} \cdot \bar{r}_t + \tau$ to obtain the upper bound in the proposition. □

The upper bound in Proposition 4.3 facilitates binary encoding of the slack variable $s_t \in [0, C_t]$ as

$$s_t^* = \frac{C_t}{2^{M_t} - 1} \sum_{m=0}^{M_t - 1} 2^m d_{t,m}, \tag{4.15}$$

where vector $d_t = (d_{t,0}, \ldots, d_{t,M_t-1}) \in \{0,1\}^{M_t}$ is a minimizer of $|s_t - s_t^*(d_t)|$, and $M_t \in \mathbb{N}$ is the number of binary variables used in the binary encoding.

Plugging in binary-encoded decision variables ξ_0^* (4.4), τ^* (4.8), y_t^* (4.12), and s_t^* (4.15) into mean-ES portfolio problem (4.2) with penalty terms (4.13) and (4.14) yields

slack-variable-based QUBO mean-ES portfolio formulation

$$
\begin{aligned}
\underset{\substack{\xi\in\{0,1\}^d,\\ a\in\{0,1\}^J, b\in\{0,1\}^K,\\ c_t\in\{0,1\}^L, d_t\in\{0,1\}^{M_t}}}{\text{maximize}} \quad & r\frac{B}{2^J-1}\sum_{j=0}^{J-1}2^j a_j + \xi\cdot\mathbb{E}[R] \\
& -\lambda\left(C+\frac{B-C}{2^K-1}\sum_{k=0}^{K-1}2^k b_k + \frac{1}{1-\alpha}\frac{1}{T}\sum_{t=1}^{T}\frac{B-C}{2^L-1}\sum_{l=0}^{L-1}2^l c_{t,l}\right) \\
& -P\left(\frac{B}{2^J-1}\sum_{j=0}^{J-1}2^j a_j + \xi\cdot\pi - B\right)^2 \\
& -\sum_{t=1}^{T}P_t\Bigg(\frac{B-C}{2^L-1}\sum_{l=0}^{L-1}2^l c_{t,l} + r\frac{B}{2^J-1}\sum_{j=0}^{J-1}2^j a_j + \xi\cdot r_t \\
& +C+\frac{B-C}{2^K-1}\sum_{k=0}^{K-1}2^k b_k - \frac{C_t}{2^{M_t}-1}\sum_{m=0}^{M_t-1}2^m d_{t,m}\Bigg)^2 .
\end{aligned}
$$

Use the LLN to estimate $\mathbb{E}[R]$ with $\frac{1}{T}\sum_{t=1}^{T} r_t$ where, as before, $\{r_1, ..., r_T\}$ are observations of risky assets return R.

4.3.2 Unbalanced-penalization approach

Let a general quadratic constraint binary optimization problem be given by

$$
\begin{aligned}
\underset{x\in\{0,1\}^b}{\text{minimize}} \quad & f(x) \\
\text{subject to} \quad & h(x) \geqslant 0,
\end{aligned}
$$

where vector $x \in \{0,1\}^b$ is the binary decision variable of the problem, objective $f(x) = x^T Q x$ is a quadratic function with an $n \times n$ real coefficient matrix Q associated to the specific problem, and constraint takes the form

$$
h(x) = D - \sum_{i=1}^{n} l_i x_i \geqslant 0,
$$

for constants $D \in \mathbb{R}$ and $l_i \in \mathbb{R}$ specific to the optimization problem, as well.

The unbalanced penalization introduces exponential decay $e^{-h(x)}$ as the penalty term. When the constraint $h(x) \geqslant 0$ holds, penalty $e^{-h(x)} \approx 0$. Otherwise, the penalty increases exponentially. QUBO formulation does not allow for exponential functions. Expand $e^{-h(x)} \approx 1 - h(x) + \frac{1}{2}h(x)^2$. The general penalty term is

$$
\zeta(x) = -\lambda_1 h(x) + \lambda_2 h(x)^2,
$$

where multipliers $\lambda_1, \lambda_2 \in \mathbb{R}$ should be tuned for the specific problem.

Apply this approach to the mean-ES portfolio problem (4.2). The function $h_t(y_t, \bar{\xi}, \tau) = y_t + \bar{\xi} \cdot \bar{r}_t + \tau$ and the penalty term is $\zeta = \sum_{t=1}^{T} -\lambda_{1,t} h_t(y_t, \bar{\xi}, \tau) + \lambda_{2,t} h_t(y_t, \bar{\xi}, \tau)^2$. Plug in the binary-encoded decision variables y_t^* (4.12), ξ_0^* (4.4), and τ^* (4.8) into ζ to obtain binary-encoded

$$h_t^*(c_t, a, \xi, b) = \frac{B-C}{2^L - 1} \sum_{l=0}^{L-1} 2^l c_{t,l} + r \frac{B}{2^J - 1} \sum_{j=0}^{J-1} 2^j a_j + \xi \cdot r_t + C + \frac{B-C}{2^K - 1} \sum_{k=0}^{K-1} 2^k b_k.$$

The full unbalanced-penalization-based QUBO formulation of the Mean-ES portfolio problem (4.2) is

$$\begin{aligned} \underset{\substack{\xi \in \{0,1\}^d, a \in \{0,1\}^J, \\ b \in \{0,1\}^K, c_t \in \{0,1\}^L}}{\text{maximize}} \quad & r \frac{B}{2^J - 1} \sum_{j=0}^{J-1} 2^j a_j + \xi \cdot \mathbb{E}[R] \\ & - \lambda \left(C + \frac{B-C}{2^K - 1} \sum_{k=0}^{K-1} 2^k b_k + \frac{1}{1-\alpha} \frac{1}{T} \sum_{t=1}^{T} \frac{B-C}{2^L - 1} \sum_{l=0}^{L-1} 2^l c_{t,l} \right) \\ & - P \left(\frac{B}{2^J - 1} \sum_{j=0}^{J-1} 2^j a_j + \xi \cdot \pi - B \right)^2 \\ & - \sum_{t=1}^{T} -\lambda_{1,t} h_t^*(c_t, a, \xi, b) + \lambda_{2,t} h_t^*(c_t, a, \xi, b)^2. \end{aligned} \tag{4.16}$$

Use the LLN to estimate $\mathbb{E}[R]$ with $\frac{1}{T} \sum_{t=1}^{T} r_t$ where, as before, $\{r_1, ..., r_T\}$ are observations of risky assets return R.

5 Case study

The case study implements (Straka 2025) the unbalanced-penalization-based QUBO (4.16) model in Python. The implementation runs the QUBO model on the quantum processing unit (QPU) at the D-Wave's Ocean SDK (software development kit). The model uses real financial data of the constituents in the S&P500 index.

5.1 Dataset

The model constructs a portfolio from S&P500 constituents. On August 1, 2025, the S&P500 index consisted of $d = 503$ constituents (Wikipedia contributors 2025). The Python code pulls daily close prices from Stooq (Stooq.com 2025). The dataset consists of the last 265 trading day prices from July 11, 2024, to July 31, 2025. The prices are adjusted for stock splits and cash dividends.

5.2 Implementation

The Basel Committee on Banking Supervision established a standard (Basel 2025) that describes the calculation of capital requirements for market risk. The standard specifies that

- the ES for market risk capital purposes is calculated based on a 12-month period (Basel 2025, p. 33.6),
- the ES is calculated at confidence level $\alpha = 0.975$ (Basel 2025, p. 33.3), and
- the ES is measured over a 10-trading-day horizon (Basel 2025, p. 33.4).

Recall that the model calculates the estimates of the ES (3.3), the covariance $\mathrm{Cov}(R)$ of returns, and the expected return $\mathbb{E}[R]$ with return observations $\{r_1, ..., r_T\}$. The Basel standard specifies that $\{r_1, ..., r_T\}$ are observations of overlapping returns over 10 trading days throughout a year.

The model follows the Basel standard. A year has approximately 255 trading days. The model calculates $T = 255$ overlapping returns $\{r_1, ..., r_T\}$ for every S&P500 constituent in the dataset.

The model assumes that the budget $B = 100$ and that each initial price π_i is the weight of the constituent i in the index S&P500 for each $1 \leqslant i \leqslant d$. This assumption makes the portfolio $\xi \in \{0, 1\}^d$ an inclusion/exclusion portfolio and minimizes the number of needed scarce qubits because the binary encoding of each ξ_i costs additional qubits.

The model further requires a calibration of the trade-off parameter λ and penalty coefficients P and P_t. This thesis does not treat the calibration of the parameters and coefficients due to time restrictions.

5.3 Results

The model outputs seemingly meaningful portfolio values $\bar{\xi}$. This thesis does not validate and calibrate the model due to time restrictions. The results cannot be interpreted as meaningful for that reason. The main result of this case study is the mere possibility of running the mean-ES portfolio on a real quantum annealer from D-Wave.

6 Discussion and future work

Prior annealing studies (Xu et al. 2023; Dasgupta and Banerjee 2020) enforce an ES threshold iteratively. These studies solve the Markowitz model on the annealer and satisfy an ES threshold by iteratively adjusting the expected return. The studies avoid direct translation of the ES onto the annealer because of a rapid growth in scarce qubits. This thesis translates the mean-ES model into the QUBO form and executes the model on D-Wave hardware. This approach should become increasingly tractable as the quantum processing unit (QPU) size and connectivity improve.

The thesis also introduces and explains the mathematical foundations of quantum annealing and its relation to QUBO optimization. As a thesis in mathematics, the theoretical background of quantum annealing and the rigorous translation from general QUBO forms to Hamiltonian operators are main parts of the contribution. In particular, the streamlined proof of Theorem 2.6 clarifies how a general QUBO model maps to a problem Hamiltonian. This translation clarifies the connection between the Hamiltonian operator conceptualization on hardware and the binary optimization model to be solved.

The main theoretical contribution of this thesis is the portfolio-independent bound on τ. This bound resolves a practical barrier in binary encoding of the mean-ES model by providing an interval for τ that does not depend on the unknown optimal portfolio. The derivation uses probabilistic concepts and is a step in making the QUBO formulation implementable on finite qubit hardware.

Methodologically, the thesis introduces two ways to eliminate inequality constraints arising in the Rockafellar and Uryasev (2000) formulation of ES. The slack-variable approach preserves feasibility via additional encoding, while the unbalanced-penalization approach reduces qubit usage through a structured quadratic approximation of an exponential penalty. These methods show the current trade-off in quantum annealing, which is between ensuring feasibility and conserving scarce qubits.

The main numerical result is the possibility of obtaining portfolio outputs from an actual quantum annealer on a realistic dataset. The implementation uses real financial data from the S&P500 constituents and follows Basel-style choices for ES estimation. The thesis emphasizes that the case study's primary outcome is feasibility, i.e., the D-Wave system runs the mean-ES QUBO using the Ocean SDK.

The thesis deliberately reduces the qubit count by assuming the inclusion or exclusion of risky assets instead of finitely discretized weights of risky assets. This assumption makes the model fit within current hardware limits, but limits the applicability of the model in various financial contexts. The anticipated development of larger and better-connected QPUs may render the assumption unnecessary and enable formulations with discretized weights.

The reformulation of ES by Rockafellar and Uryasev (2000) yields a linear program

(LP) (3.5). Classical solvers effectively solve LPs. Showing an advantage in using QPUs for solving LPs is presumably difficult. This observation is consistent with the piecewise linear structure of the mean-ES objective. Augmentation of the model with nonconvex, integer-based, or quadratic terms poses a greater challenge for classical solvers. The quantum advantage may be easier to demonstrate on such augmented models. The mean-ES model in QUBO format presented in this thesis provides a foundation for such augmented models. In addition, the use of ES aligns with the Basel III risk-measure framework, which motivates the focus on this coherent tail-risk criterion.

The thesis translates a mean-ES portfolio to QUBO and executes the QUBO model on a D-Wave annealer. The tuning of the penalty P for violating the budget constraint and coefficients $\lambda_{1,t}$, $\lambda_{2,t}$ in the unbalanced penalization remains. A study identifying optimal values of these parameters remains future work. Preliminary tuning suggests that the QPU can output non-trivial inclusion/exclusion portfolios rather than degenerate solutions that effectively invest into a single asset, but the thesis refrains from assigning financial meaning to these outcomes without a systematic calibration and validation study.

The thesis model calculates the expected shortfall using observations in the tail of the loss distribution. Since there are few such observations, the ES-based portfolio is expected to be sensitive to sampling variation. At the same time, the present estimation of ES does not impose a distributional assumption on losses, and the use of ES aligns with Basel-style regulatory practice that prioritizes tail-risk sensitivity.

Xu et al. (2023) suggests another approach for incorporating ES into QUBO. Convert ES to a multilinear polynomial through Fourier analysis (O'Donnell 2014) and then convert the multilinear polynomial to a quadratic polynomial with quadratization (Dattani 2019; Verma, Lewis, and Kochenberger 2021; Mandal et al. 2020). Future work may compare the qubit usage of this approach in comparison to the approach in this thesis.

The following list suggests future work directions.

1. **Validation and scalability.** Run the model on small benchmark problems with known solutions and gradually scale to larger instances to confirm model correctness and study how solution quality evolves on hardware.

2. **Hyperparameter calibration.** Systematically calibrate λ, P, and the unbalanced-penalization coefficients $\lambda_{1,t}$ and $\lambda_{2,t}$ (1) to quantify the trade-offs between feasibility (satisfying the budget and constraints) and quality (obtaining non-trivial portfolios with good mean-ES performance) and (2) to improve stability across runs.

3. **Robust ES estimation.** Measure the robustness of the ES component under alternative window lengths, larger datasets, or alternative estimators that may reduce tail-noise without introducing strong distributional assumptions.

4. **Augmented objective functions.** Extend the objective toward genuinely quadratic or more combinatorial structures to create quantum-versus-classical comparisons.

5. **Discretized weights.** When qubit resources suffice, extend the inclusion/exclusion portfolio toward discretized weights. The present restriction is a deliberate choice to enable execution on real hardware within a small qubit count.

6. **Dynamic portfolios and additional frictions.** Explore multi-period versions of the model and incorporate practical terms such as transaction costs and other institutional constraints.

7. **Tighter variable bounds.** Derive tighter portfolio-independent bounds for auxiliary variables (including τ) to further reduce binary-encoding overhead and improve scalability.

In summary, this thesis provides a mathematical framework for translating mean-ES optimization to quantum annealing and a demonstration of real-hardware implementation on S&P500 data. The current results should be seen as a feasibility milestone and a theoretical foundation. The main parts are the bound on τ, the clarified QUBO-to-Hamiltonian proof, and the two constraint-elimination strategies. These results may support more financially relevant models in future work.

Popular Summary

Financial markets are usually calm, but sometimes they fall fast. A portfolio that looks fine in average conditions can still suffer large losses in rare downturns. This thesis asks whether a modern way of measuring that downside risk can be implemented with today's quantum annealers, which are types of a quantum computer solving optimization problems.

The work has two parts. The first part explains quantum annealing and the problem form that current devices accept. The second applies this quantum annealing to portfolio optimization with Expected Shortfall (ES), which is a tail-risk measure used in modern regulation. The main message is that this tail-risk model can be rewritten in a form an annealer can run.

A helpful intuition behind quantum annealing is a landscape of hills and valleys. Each assignment of many binary variables (0 or 1) corresponds to one point in the landscape, and the height of that point is the cost (often called the energy). The solver's aim is to find a deep valley. A quantum annealer builds this landscape using qubits and tries to move toward a low-energy state.

The thesis introduces the minimal quantum ideas needed to support this picture. The device starts from a simple quantum system whose lowest-energy state is easy to prepare. The device then changes the system gradually so that, at the end, the energy of each 0/1 configuration matches the cost of the optimization problem. Measuring the 0/1 qubit values at the end produces a candidate optimization solution.

For real devices, the optimization model must be written as a QUBO (quadratic unconstrained binary optimization). A QUBO uses binary variables and an objective function with only linear and quadratic terms. The thesis explains how the QUBO coefficients become interactions between qubits.

With this quantum background in place, the thesis turns to portfolio risk. The classical Markowitz mean-variance model balances expected return against variance. Because variance already gives a quadratic objective, mean-variance fits QUBO straightforwardly and has become a common research topic in quantum finance. But variance does not focus on the worst losses. Expected Shortfall does. ES averages the losses in the worst tail beyond a chosen confidence level.

A reformulation of ES introduces an auxiliary variable τ and uses terms of the form $\max(0, L - \tau)$. This makes ES more compatible with a quadratic binary model than many other advanced risk measures. Turning a Mean-ES portfolio into QUBO raises three practical issues. First, continuous quantities must be approximated with binary expansions. Second, the inequality constraints from the ES reformulation and the budget rule must be enforced with QUBO penalties; the thesis compares slack-variable and unbalanced-penalty approaches. Third, binary encoding requires bounds, and the thesis

derives a portfolio-independent bound for τ.

Finally, the thesis demonstrates feasibility on S&P 500 data in a Basel-style setting. The implementation uses an inclusion/exclusion portfolio rather than finely discretized weights to fit current hardware limits. The QUBO is implemented with D-Wave's Ocean tools and run on a quantum processing unit. The numerical study is presented as a feasibility test rather than a fully tuned investment method.

In summary, this thesis links quantum annealing theory with a tail-risk portfolio model that matters in practice. It shows how QUBO acts as the connection between portfolio optimization and quantum annealing hardware and how Expected Shortfall can be reshaped to fit that QUBO connection. The result is a clear route from a realistic risk objective to an executable QUBO on current annealers, and a possible starting point for more advanced models as hardware improves.

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